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Predictive Analytics for Enhancing Sustainability in Petroleum Operations

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By

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UNDERTAKING

This is to declare that the project entitled "**Predictive Analytics for Enhancing Sustainability in Petroleum Operations**" is an original work done by the undersigned, in partial fulfillment of the requirements for the degree Bachelor of Science at Computer Science, College of Computer Science and Information technology King Faisal University.

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ABSTRACT

The significant contribution of CO_2 emissions to global climate change created a need for innovative solutions in emissions monitoring and forecasting. This project introduces an Internet of Things (IoT)-enabled system designed to measure real-time CO_2 emissions and integrate these data into a Hybrid learning framework for forecasting future emissions. Utilizing both Deep learning (DL) and Machine learning (ML) techniques such as Long Short-Term Memory (LSTM) networks and Extreme Gradient Boosting (XGBoost), the system analyzes historical and real-time data to identify patterns and trends. The IoT device leverages precise sensors to ensure accurate emission tracking, which supports both operational optimization and compliance with environmental regulations. This integrated approach enables proactive decision-making, enhancing resource management and aligning with sustainability goals. Designed with scalability and accessibility, the system provides actionable insights through an intuitive interface, bridging technical and non-technical stakeholder needs.

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1 Introduction

1.1 Background

The oil industry plays a critical role in the global energy supply chain, driving economic growth and supporting various sectors.[1] Nevertheless, it also serves as a significant contributor to CO_2 emissions, which are key drivers of climate change and environmental degradation [1]. This industry is responsible for substantial carbon dioxide (CO_2) and methane emissions stemming from extraction, refining, and transportation activities.

Considering the increasingly noticeable effects of climate change, the oil sector is encountering mounting pressure from governments, regulators, and society to embrace more environmentally sustainable practices. Stringent environmental policies coupled with the worldwide transition to cleaner energy sources are compelling industry to reassess its methods. Consequently, sustainability has emerged as a paramount concern, with predictive analytics presenting a promising avenue for addressing environmental issues in industry by curbing emissions and enhancing resource efficiency.[2] Through the analysis of historical data, predictive analytics empowers companies to foresee potential future environmental repercussions, thereby facilitating more informed and effective decision-making processes. With the advancement of machine learning (ML) technologies, the oil industry can harness these tools to bolster sustainability initiatives while maintaining operational excellence. Predictive models furnish actionable insights that aid the industry in reducing its environmental impact and steering towards more sustainable business practices.

The Oil Industry faces a lot of challenges during extraction, production, and distribution processes in the oil sector resulting in significant emissions, resource depletion, and ecological disruption. These challenges have become a major concern for policymakers and industry stakeholders. With the help of Predictive analytics, it can provide valuable insights by forecasting emissions, operational efficiency, and identifying optimization opportunities. By leveraging historical data, assist in proactive decision-making for sustainable practices.

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[1] *Investopedia. (2024). What percentage of the global economy is the oil and gas drilling sector?*
[2] *Haddadian, R. (2024,). How real-time machine learning can help with emissions monitoring.*

By using both Machine Learning (ML) and Deep Learning models both are particularly well-suited for handling complex datasets from the oil industry. Both can identify patterns in emissions, production metrics, and operational efficiency, aiding in the development of sustainable strategies.

1.2 Motivation

The petroleum industry is a cornerstone of global energy supply, but it also represents one of the largest contributors to CO_2 emissions. With increasing environmental and regulatory pressures, the industry faces significant challenges in adopting sustainable practices.[3] However, existing predictive analytics systems/models in this domain often fall short due to several limitations.

First, Refining is the most energy consuming process utilized in practice. Crude oil refineries belong to the largest consumers of utilities in process industries [4]. Therefore, Complex distillation-based operations are the first to be mentioned regarding the efforts for short and long term [2]. Even though energy saving is accompanied by a corresponding reduction in emissions, there is an increasing need for explicit estimation of the emissions associated with various energy-intensive refining-based operations in process industries.

Second, Forecasting Accuracy in Traditional ML and Deep Learning Approaches. The traditional machine learning models, like linear regression, cannot capture the complex nonlinear dynamics of CO_2 emissions. These models struggle to capture the intrinsic dependencies among variables such as operational metrics, environmental conditions, and emission output. This limits their prediction accuracy, making them less effective for decision-making in sustainability applications. Advanced methods can overcome it by leveraging temporal patterns of emission data. Examples are LSTMs that could capture the sequential dependencies from their historical data to give forecasts that are far more reliable and accurate. It provides a significant enhancement to predictive capabilities in identifying trends and mitigating against any undue emissions [5].

[3] *Investopedia. (2024). What percentage of the global economy is the oil and gas drilling sector?*

Third, User Accessibility Issue. Most emissions monitoring systems lack intuitive, easy-to-use interfaces that would be required for effective adoption by non-technical stakeholders. This has confined their utility in critical decision-making processes where ease of access and clear data presentation is particularly important. For instance, overly complex dashboards and poorly laid-out visualizations can inhibit users from making informed decisions with the help of emissions data or using predictions. Accessibility improvement will be in doing interfaces with simplified workflows, clear visual representations, and customized dashboards, catering to various user needs. Adding more features in, such as role-specific functionality for both technical and non-technical users, can better drive adoption and operational efficiency in the gap between data complexity and user capability.

After analyzing these drawbacks, we can identify the following key challenges that need to be addressed for effective emissions management systems:

1. **Prediction Accuracy**
    - Traditional machine learning models, such as regression, fail to capture the complex nonlinear dynamics of CO_2 emissions.
    - These models do not account for intrinsic dependencies among variables like operational metrics, environmental conditions, and emission outputs, leading to inaccurate forecasts.
2. **User Accessibility**
    - Emissions monitoring systems lack intuitive, easy-to-use interfaces, making them difficult for non-technical stakeholders to adopt.
    - Overly complex dashboards and poor visualizations impede effective decision-making based on emissions data.
3. **Wasting resources**
    - Refining is the most energy-intensive process in the petroleum industry, making it a significant contributor to overall energy demand in process industries.

1.3 Problem Statement

The petroleum refining sector faces mounting sustainability challenges, particularly in managing CO_2 emissions. A primary issue lies in the underutilization of resources and the unpredictable nature of emissions, which hinder companies' ability to proactively address environmental impacts [6].

Furthermore, the sector's operational complexity complicates the balance between maintaining efficiency and adhering to increasingly stringent environmental regulations [7]. Companies may lack predictive tools that can anticipate emission trends, leading to missed opportunities for risk reduction and process optimization.

This project aims to bridge this gap by developing a predictive analytics framework specifically for the oil refining sector. Using historical data, the system will forecast emissions, optimize resource utilization, and ensure compliance with environmental regulations. By providing accurate, data-driven insights, this project will empower companies to reduce their carbon footprint, improve sustainability, and enhance decision-making in a manner that is both proactive and efficient.[4]

1.4 Scope of the Project

The oil sector remains a vital part of the global economy but is also one of the largest contributors to greenhouse gas emissions,[5] driving environmental degradation and climate change. As sustainability becomes a global priority, companies are under increasing pressure from regulators, stakeholders, and the public to minimize their ecological footprint [8].

This project proposes the development of a predictive analytics system to address these challenges. Using a combination of Machine learning techniques and Deep learning resulting in Ensemble Learning methodology (Hybrid Learning). The system will analyze historical emissions data and operational parameters to identify patterns and forecast future emissions.

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[4] *Haddadian, R. (2024). How real-time machine learning can help with emissions monitoring.*
[5] *U.S. Environmental Protection Agency. (2014). Emission factors for greenhouse gas inventories.*

Designed for adaptability, the system will be capable of accommodating diverse operations across the oil sector, making it flexible for varying contexts and requirements. Additionally, it will improve the accuracy of emissions forecasting and help identify the key factors influencing environmental performance. By linking data analysis to actionable insights, this project aims to provide oil companies with a comprehensive tool to implement effective and measurable strategies for reducing emissions and achieving sustainability.

1.5 Project Innovation and Utilities

The oil industry, as a significant contributor to global greenhouse gas emissions, faces urgent challenges related to climate change and pollution.[6] Despite increased awareness of these issues, companies often struggle to implement effective sustainability measures. Traditional methods rely on periodic assessments that lack real-time insights into operational impacts, leaving companies unable to respond effectively to environmental challenges. There is an urgent need for innovative systems that assist stakeholders in tracking, analyzing, and reducing their environmental footprint.

This project introduces a novel approach by leveraging predictive analytics and machine learning to provide oil companies with a data-driven framework for addressing these challenges. By analyzing historical emissions data and operational metrics, the system will identify potential environmental risks and recommend actionable strategies for emissions reduction. This predictive model will support compliance with environmental regulations while enhancing operational sustainability. The innovation lies in its ability to transform raw data into actionable insights, enabling companies to proactively manage their emissions, optimize resources, and adopt sustainable practices. This solution not only benefits oil companies but also contributes to broader global efforts to combat climate change and reduce industrial emissions.

[6] *International Energy Agency. (2023). Emissions from Oil and Gas Operations in Net Zero Transitions*

1.6 Expected Outcomes

The primary objective of this project is to contribute to environmental sustainability by addressing the challenges posed by emissions in the oil industry. It aims to develop solutions that enhance efficiency, reduce environmental impact, and support sustainable practices, ensuring a balance between industrial operations and ecological responsibility. Specifically, the project aims to:

1. **Develop a Predictive Analytics System**

   Improve accuracy in predicting emissions and enabling proactive decision-making to minimize environmental impact. The system leverages historical data and machine learning models to accurately forecast greenhouse gas emissions. This enables stakeholders to anticipate future sustainability challenges and take proactive measures to mitigate environmental harm. By adopting predictive analytics, the system aligns with initiatives like the International Energy Agency's recommendations for integrating data-driven tools in emission reduction strategies.[7]

2. **Optimization of Resource Management**

   Identify inefficiencies and recommend strategies for sustainable resource allocation. The platform analyzes operational metrics to detect resource wastage and proposes actionable recommendations to enhance efficiency. By optimizing resource allocation, companies can reduce costs and improve operational sustainability. This outcome supports advancements in industrial efficiency outlined in studies on resource optimization in oil and gas operations.

[7] *international Energy Agency. (2023). Emissions from Oil and Gas Operations in Net Zero Transitions*

3. **Create a User-Friendly Platform**

Design an intuitive interface with real-time dashboards for easy navigation and data input. The platform is designed with usability in mind, featuring a clean and intuitive interface that accommodates users with varying technical expertise. Real-time dashboards and easy data input methods ensure accessibility and encourage widespread adoption of sustainable practices among stakeholders.

4. **Improve Decision-Making for Stakeholders**

Provide actionable analytics and visualization tools for informed strategic planning. The system empowers decision-makers by offering comprehensive analytics and real-time visualizations. This enables stakeholders to understand complex operational data briefly, fostering informed decisions that balance operational goals with sustainability objectives

1.7 Stakeholders and Their Roles

The primary stakeholders for this project encompass key groups and entities that play essential roles in ensuring the success of the system, its implementation, and its contribution to sustainability goals. These stakeholders come from various sectors. Below is an elaboration of each stakeholder group:

- **Oil Companies**

  Directly benefiting from its tools and insights. Decision-makers leverage the system to monitor emissions, anticipate future trends, and align operations with environmental and business goals. Operational teams engage with the system to input data, track performance metrics, and implement actionable recommendations that enhance resource management and reduce emissions. Compliance teams ensure adherence to local and international environmental regulations using predictive insights and reporting tools, positioning oil companies as central players in the industry's sustainability transition.

- **Regulatory Bodies**

  Ensuring that oil companies comply with environmental laws and standards. Environmental monitors utilize the system to track compliance and identify areas needing improvement. Policy developers analyze the system’s data insights to craft regulations that address emerging sustainability challenges. Auditors validate emissions reports generated by the system to ensure accuracy and accountability, promoting a culture of transparency and environmental responsibility within the industry.

- **Technology Developers (Internal or External)**

  Play a crucial role in creating and maintaining the system. System developers ensure the platform remains functional, user-friendly, and capable of meeting evolving stakeholder needs. Data scientists refine predictive analytics models, enhancing their accuracy and reliability for emissions forecasting. Support teams provide training and technical assistance, ensuring seamless adoption and effective usage of the system across different user groups.

- **Communities Affected by Oil Operations**

  Communities impacted by oil operations represent the societal beneficiaries of the project. Improved air quality and reduced environmental degradation are direct benefits they experience due to the system's implementation. Watchdog groups within these communities use system-generated public reports to hold companies accountable for minimizing their environmental impact, fostering transparency and accountability.

- **Research Institutions**

  Contribute through expertise in refining data collection methods, improving predictive models, and developing emissions mitigation strategies. As knowledge providers, they collaborate with oil companies and regulators to ensure the system is based on sound scientific principles. Evaluators assess the system's effectiveness and propose enhancements, ensuring continuous improvement and alignment with global sustainability efforts.

1.8 Environmental Impact

The project holds significant potential to positively impact the environment by addressing the pressing challenges associated with emissions in the oil industry. These impacts are both direct, through tangible reductions in greenhouse gas emissions, and indirect, by fostering systemic change within industry practices and stakeholder collaboration. Below is an elaboration of these environmental impacts:

1. **Optimization of Resource Use**

   By analyzing operational metrics, the system helps companies reduce waste and utilize resources more efficiently. This results in a lower environmental footprint, conserving natural resources, and minimizing the ecological impact of oil production and refining processes.

2. **Support for Proactive Compliance with Environmental Regulations**

   The system forecasts emissions trends, enabling companies to anticipate and comply with evolving environmental regulations. This proactive approach not only reduces the likelihood of regulatory violations but also fosters a culture of accountability and environmental stewardship within the industry.

3. **Promotion of Industry-Wide Sustainability**

   The project encourages collaboration among stakeholders, including oil companies, regulatory bodies, and environmental organizations. This collective effort fosters the adoption of innovative practices and technologies that improve the oil industry's overall sustainability and alignment with global environmental goals.

2 Literature Review

Predictive analytics and machine learning have been recognized for their ability to improve sustainability and operational efficiency in the oil and gas sector. By processing complex datasets and forecasting outcomes with high accuracy, these tools offer significant potential for enhancing industry practices.

By using and leveraging the power of Deep learning models and machines. Learning we can achieve more accurate predictions.

2.1 Relevant Studies

The petroleum industry faces increasing challenges to balance operational efficiency with environmental sustainability. Predictive analytics and machine learning have emerged as vital tools to address these issues, offering advanced methods to analyze data, forecast emissions, and optimize processes. This section reviews key studies that highlight innovative applications of AI and predictive models in reducing CO_2 gas emissions and improving resource management.

1. **Application of Machine Learning in Methane Emissions Modeling**

   This study addresses methane emissions in the petroleum industry, a key contributor to greenhouse gases. Using Long Short-Term Memory (LSTM) networks, the research models temporal patterns of methane emissions by leveraging time-series data from oil production and refining activities. The model achieves a high accuracy of 93% and effectively captures sequential dependencies. While it provides critical insights for managing methane emissions, the model's scalability to other pollutants is limited [5].

2. Application of Artificial Intelligence to Predict CO_2 Emissions

This study explores the use of multiple artificial intelligence models—Long Short-Term Memory (LSTM), Feedforward Neural Networks (FFNN), and Adaptive Neuro-Fuzzy Inference System (ANFIS)—to predict CO_2 emissions. The ensemble approach leverages the strengths of these models, achieving high predictive accuracy, with LSTM outperforming the others (RMSE: 15.42, R^2: 0.995). However, the study highlights the need for extensive computational resources and dependency on high-quality input data, which may limit its broader applicability [9].

Machine Learning-Based Time Series Models for Effective CO_2 Emission Prediction in India

This study focuses on CO_2 emission prediction specific to India, comparing models such as Artificial Neural Networks (ANN), Random Forest (RF), and Long Short-Term Memory (LSTM). LSTM demonstrated the highest accuracy (MSE: 0.007, R^2: 0.94) in forecasting emissions. The study provides valuable insights into selecting effective models for time-series data. Nevertheless, model performance is sensitive to dataset variations, necessitating careful model selection and tuning for optimal results [10].

3. **Predicting CO_2 Emission Footprint Using AI through Machine Learning**

This research evaluates multiple machine learning algorithms—Multiple Linear Regression (MLR), Support Vector Machine (SVM), and Random Forest (RF)—to predict CO_2 emissions. Among these, SVM achieved the highest precision (MAE: 0.198, RMSE: 0.245). The study offers a comprehensive comparison of these models, showcasing their applicability in emissions prediction. However, the results underline the importance of careful feature selection and dataset-specific optimizations to maintain predictive performance [11].

4. **Assessment of Advanced Machine and Deep Learning Approaches for Predicting CO_2 Emissions from Agricultural Lands**

This study examines the performance of several models, including Feedforward Neural Networks (FNN), Convolutional Neural Networks (CNN), Gradient Boosting Regression (GBR), and Support Vector Regression (SVR), in predicting CO_2 emissions in agricultural contexts. FNN emerged as the most accurate model (R^2: 0.918, RMSE: 67.75). While the study offers a detailed comparison of machine and deep learning models, it highlights the challenges of achieving consistent performance across diverse datasets, emphasizing the need for careful model selection and tuning [12].

Summary of Relevant Studies

Table 1 Relevant Studies Summary

Study Title	Authors	System Model	Accuracy	Advantages	Limitations
Application of ML in Methane Emissions Modeling	R. Luo	(LSTM)	(MAE): LSTM: 0.254	Effective for temporal and sequential data high accuracy.	Limited scalability to other pollutants.
Application of Artificial Intelligence to Predict CO_2 Emissions.	Semen Budennyy, Vladimir Lazarev, Nikita Zakharenko	(FFNN) (ANFIS)	(RMSE): FFNN: 19.78 ANFIS: 20.90 (R^2): FFNN: 0.991 ANFIS: 0.989	High predictive accuracy; ensemble approach leverages strengths of multiple models.	Requires extensive computational resources; dependent on quality of input data.
Machine Learning-Based Time Series Models for Effective CO_2 Emission Prediction in India	Xiangqian Li	(ANN, LSTM)	(MSE): ANN: 0.010 LSTM: 0.007 (RMSE): ANN: 0.100 LSTM: 0.084 (MAPE): ANN: 1.95% LSTM: 1.65% (R^2): ANN: 0.90 LSTM: 0.94	Identifies most effective models for CO_2 emission prediction; provides insights specific to India's emissions.	Model performance may vary with different datasets; requires careful model selection.
Predicting CO_2 Emission Footprint Using AI through Machine Learning	Yang Meng, Hossain Noman	(MLR), (SVM)	(MAE): MLR: 0.254 SVM: 0.198 (MAPE): MLR: 5.12% SVM: 3.98% (RMSE): MLR: 0.312 SVM: 0.245	Comprehensive comparison of different machine learning models; identifies SVM as achieving the highest precision in CO_2 emission predictions	Performance may vary with different datasets; requires careful feature selection
Assessment of Advanced Machine and Deep Learning Approaches for Predicting CO_2 Emissions from Agricultural Lands	A. K. Mishra, S. K. Singh, P. K. Singh	(FNN) (CNN) (GBR)	(R^2): GBR: 0.887 CNN: 0.886 (RMSE): GBR: 79.88 CNN: 80.11 (NSE): GBR: 0.880 CNN: 0.880	Comprehensive comparison of machine and deep learning models; identifies FNN as the most accurate for CO_2 emission predictions in agricultural contexts.	Performance may vary with different datasets; requires careful model selection and tuning.

2.2 Identified Gaps and Proposed Achievements

Despite advances in predictive analytics for emissions management, the following gaps remain unaddressed:

Identified Gaps of Studies

Table 2 Identified Gaps

Identified Gap	Studies Sharing the Gap	Reference Numbers
Limited scalability to address emissions from multiple pollutants or external factors.	Machine Learning-Based Time Series Models	[10]
High dependency on computational resources and high-quality input data.	Application of AI to Predict CO_2 Emissions, Assessment of Advanced Machine and Deep Learning Approaches	[9], [12]
Performance sensitivity to dataset variations, requiring extensive tuning and careful model selection.	Machine Learning-Based Time Series Models, Assessment of Advanced Machine and Deep Learning Approaches	[10], [12]
Heavy reliance on feature selection and dataset-specific optimizations for performance.	Predicting CO_2 Emission Footprint Using AI, Application of AI to Predict CO_2 Emissions	[11], [9]
Limited generalizability to diverse operational or environmental contexts.	Assessment of Advanced Machine and Deep Learning Approaches, Predicting CO_2 Emission Footprint Using AI	[12], [11]

Proposed Contributions

Aim to address the gaps identified in each study by offering solutions that align with the objectives of enhancing sustainability in petroleum operations

Table 3 Proposed Contributions

Identified Gaps	Identified contribution
[11]	Develop easy-to-use tools for emissions prediction with real-time applications. Integrate predictions into actionable strategies for operational improvements. Use predictions to reduce resource wastage in emissions-heavy operations.
[12]	Use predictive models to optimize resource allocation in agriculture while minimizing emissions.
[5]	Provide actionable insights and real-time visualizations to manage emissions effectively.
[10]	Refine models to handle time-series data and incorporate external factors like business and/or sustainability policies. Provide recommendations based on emissions data trends for more effective planning. Use accurate forecasts to support proactive mitigation strategies in emissions-heavy industries.
[9]	Leverage AI to improve resource allocation by predicting high-emission scenarios in advance. Use predictive insights to align operations with broader sustainability goals. Simplify the computational process to make the system accessible for broader applications.

3 Preliminary Solution Identification and Justifications

3.1 Dataset Description

Two datasets will be utilized: numbering dataset names

1. **Saudi Central Bank, Oil and Gas Statistical Reports**: Provides operational metrics for oil production.[8]

Table 4 Saudi Central Bank, Oil and Gas, Features

Feature Type	Features Name	Elaboration
Categorical Features	Year	Represents the time dimension, crucial for trend analysis. It is treated categorically as each year is unique.
Independent Features	Total Production, Percentage Change, Daily Average, Fuel Oil, Diesel, Gasoline, Naphtha, LPG, Kerosene Aviation Fuels, Asphalt & Others, Coke, Fuel Gas	These variables represent crude oil production metrics and refined product outputs. They act as predictors for understanding variations in the dependent feature.
Dependent Feature	Total Refined Products	Represents the aggregated output of all refined products. This is the key target variable influenced by independent features.

[8] *Saudi Central Bank*. Oil and Gas Statistical Reports. *https://www.sama.gov.sa/en-US/EconomicReports/Pages/report.aspx?cid=126*

2. **Our World in Data (OWID) World Emissions Dataset**: Contains historical records on CO_2 emissions and their sources. [9]

Table 5 Our World in Data Features

Feature Type	Features Name	Description
Categorical Features	country, year	Country represents the geographic location for the observation, providing regional context. Year represents the temporal aspect of the data, linking trends in emissions and energy usage over time.
Independent Features	oil_co2, gas_co2, coal_co2, and other continuous variables in the dataset with a direct relationship to emissions (e.g., cement_co2, flaring_co2, etc.)	These variables are continuous metrics capturing CO_2 emissions from various sources (oil, gas, coal, cement, flaring, etc.). They directly indicate emissions linked to specific energy production or usage processes and are critical for understanding their contribution to overall emissions.
Dependent Feature	total_co2 or other aggregate metrics such as co2_per_capita	Represents total CO_2 emissions or normalized versions (e.g., emissions per capita). These act as target variables influenced by the independent features, providing insights into overall or per-unit contributions to emissions.

3.2 Data Transformation

To prepare the data for machine learning model training, several preprocessing techniques will be applied:

- **Normalization:** This technique will ensure that all features are on the same scale, particularly important when units differ (e.g., emissions measured in tons and energy in kilowatt-hours).
- **Feature Selection:** Key operational metrics that have the most predictive power in emissions forecasting will be identified and focused upon.
- **Handling Missing Data:** Imputation methods will be utilized to ensure data completeness, allowing the ML algorithms to function optimally.

[9] *CO_2 and Greenhouse Gas Emissions. Our World in Data.* https://ourworldindata.org/co2-and-greenhouse-gas-emissions

3.3 Algorithms

To build a robust model, we will use Ensemble learning (Hybrid Learning) , a technique where multiple Learners (models) are trained and their predictions are combined to improve the overall performance of the system [13]. This method leverages the strengths of individual models to compensate for their weaknesses, resulting in enhanced accuracy and robustness compared to single-model approaches [14]. For this system, we chose Long Short-Term Memory (LSTM) and Extreme Gradient Boosting (XGBoost).

Long Short-Term Memory (LSTM) networks The LSTM is a powerful recurrent neural system specially designed to overcome the exploding / vanishing gradient problems that typically arise when learning long-term dependencies, even when the minimal time lags are very long. The algorithm main usage falls into NLP (natural language processing) or Timme Series, the idea of this algorithm that it relates to processing the information they receive from previous neuron and applying the activation function from scratch [15].

LSTM will be employed to forecast emissions by analyzing time-series data such as historical CO_2 emissions and operational metrics. Its ability to maintain information over extended time intervals makes it particularly effective for predicting trends and identifying patterns in complex temporal datasets [16]. LSTM is well-suited for modeling the dynamic and sequential nature of emissions data, providing accurate forecasts that assist in proactive decision-making [5].

Extreme Gradient Boosting (XGBoost) or Tree boosting is a highly effective supervised and widely used machine learning algorithm that builds on gradient boosting principles, Where it uses a more optimized way to implement a tree-based algorithm, XGBoost uses a Greedy approach for the building of its tree, and it is a simple way to optimize the algorithm [17].

XGBoost will be utilized to predict emissions and operational efficiency by analyzing structured datasets, such as production levels and energy usage. By leveraging advanced techniques like regularization and parallel processing, XGBoost achieves high predictive accuracy and robustness. Additionally, it ranks feature importance, enabling the identification of critical factors contributing to emissions, making it invaluable for optimizing resource management and sustainability planning.

3.4 Model Training

Long Short-Term Memory (LSTM):

The training starts with, begins with data preprocessing, where the data is rescaled. The normalization will help the model to converge faster and avoid issues. Time series data is then split into windows, where each window consists of past observations used to predict the next step. The model architecture will then be defined, the number of hidden layers, the number of neurons in each hidden layer, the input layer and the output layer. For compilation, an optimization process will be implemented like Gradient Descent to optimize the training process. Loss function will be implemented to handle outliers.

During the training a data batch size will be determined with the number of epochs, which is the number of complete passes through the training data set. To ensure efficient training and prevent overfitting. Early stopping mechanism will be implemented. Which is monitoring a metric and stopping the training when there is no improvement observed.

Finally, we will use Hyperparameter tuning. implementing the most appropriate function will be defined later. Hyperparameter tuning ensures the time series nature of the data is respected. This will help identify the optimal values for model Hyperparameters.

XGBoost:

After the data normalization process is complete, we split out data into training and testing sets (e.g. 80% training and 20% testing). Then we set up the XGBoost model by defining key parameters like number of trees, learning rate, tree depth, the fraction of data used to build the tree and the proportion of features used for each tree.

For the evaluation of this model, time cross-validation will be used rather than the normal k-fold cross-validation to ensure that the model respects the time order of the data.

Finally, to fine-tune the model, hyperparameter tuning will be performed using methods that will be defined later based on the model itself (e.g. GridSearchCV or RandomizedSearchCV),

which will search for the best combination of hyperparameters while respecting the time series nature of the data.

3.5 Model Evaluation

For these models, two different evaluation functions will be used. Using more than one evaluation metric in model assessments provides a comprehensive understanding of model performance across various aspects. Each metric offers unique insights, reducing the risk of biased evaluations and improving decision-making [18].

The first evaluation metric is Mean Absolute Percentage Error (MAPE) is a widely used metric for evaluating the prediction accuracy of models [18]. It calculates the mean of absolute percentage differences between actual and predicted values, expressed as a percentage of the actual value. This evaluation

Formula 1:

$$MAPE = \frac{1}{n}\sum_{i=1}^{n} \left|\frac{yi - \hat{y}i}{yi}\right| \times 100$$

where:

n**:** The total number of observations in the dataset.

yi**:** The actual value for the i - th observation.

$\hat{y}i$**:** The predicted value for the i - th observation.

$|yi - \hat{y}i|$**:** The absolute difference between the actual and predicted values for each observation.

$\left|\frac{yi - \hat{y}i}{yi}\right|$**:** The percentage error for each observation, scaled by the actual value.

$\frac{1}{n}\sum_{i=1}^{n}$: The mean of the percentage errors across all observations.

Symmetric Mean Absolute Percentage Error (sMAPE) is a popular evaluation metric for predictive models, particularly in time-series forecasting. It measures the accuracy of predictions by comparing the absolute difference between actual and predicted values, normalized symmetrically by the average of those values. This normalization addresses the inherent bias in standard MAPE when actual or predicted values approach zero.[10]

Formula 2:

$$sMAPE = \frac{1}{n}\sum_{i=1}^{n} \frac{|yi - \hat{y}i|}{(|yi| + |\hat{y}i|)/2} \times 100$$

nThe total number of observations in the dataset.

yi: The actual value for the i-th observation.

$\hat{y}i$: The predicted value for the i-th observation.

$|yi - \hat{y}i|$**:** The absolute difference between the actual and predicted values for each observation.

($|yi| + |\hat{y}i|)/2$: The symmetric average of the absolute values of the actual and predicted values, which serves as the denominator.

$\frac{|yi - \hat{y}i|}{(|yi|+|\hat{y}i|)/2}$: The normalized symmetric error for each observation.

$\sum_{i=1}^{n}$: The mean of the normalized symmetric errors across all observations.

The reason for choosing these two-evaluation metrics is that these two are less sensitive to outliers and ideal to use where there are little to no zero values, and if the data set is not sparse.

[10] *How to Estimate Forecasting Quality: A System-Motivated Derivation of Symmetric Mean Absolute Percentage Error (SMAPE) and Other Similar Characteristics" (2014). Departmental Technical Reports* *https://scholarworks.utep.edu/cs_techrep/865*

3.6 System Hardware Overview

The Raspberry Pi 5

provides a good balance of processing power, connectivity, and affordability for IoT-based predictive analytics. It can handle sensor data collection, run basic models locally, and connect easily to cloud services, all in a compact form. This makes it a practical choice that simplifies implementation and helps keep project costs down.

Table 6 Raspberry Pi Specs

Feature	**Details**
Processor	2.4GHz quad-core Arm Cortex-A76 CPU for enhanced performance
Graphics	Video Core VII GPU supporting OpenGL ES 3.1 and Vulkan 1.2, with dual 4Kp60 output via micro-HDMI
Memory	Available in 2GB, 4GB, and 8GB LPDDR4X-4267 SDRAM configurations
Connectivity	Dual-band 802.11ac Wi-Fi, Bluetooth 5.0/BLE, and Gigabit Ethernet with PoE+ support (via PoE+ HAT).
USB Ports	2 × USB 3.0 (5Gbps) and 2 × USB 2.0 for versatile connectivity.
Expansion Interfaces	2 × 4-lane MIPI camera/display transceivers and a PCIe 2.0 x1 interface for high-speed peripherals.
Performance	Delivers 2-3x performance improvement over Raspberry Pi 4 due to upgraded CPU and GPU

Adafruit SCD-30

provides reliable CO2 measurements, along with temperature and humidity readings if needed. Its I2C interface and strong documentation simplify integration with the Raspberry Pi, reducing development time and ensuring accurate CO2 data for effective predictive analytics focused on emissions.

Table 7 Adafruit SCD-30, Specs and Features

Topic	**Key Details**
Features	Measures CO2, temperature, and humidity using NDIR technology. Suitable for environmental sensing and air quality studies.
Pinout Details	Accurate CO2 readings (ppm), integrated SHT31 sensor for temperature and humidity, 3.3V/5V compatibility, I2C interface.
Arduino Integration	- VIN: Accepts 3-5V input. - SCL & SDA: I2C clock and data pins with 10k pull-ups. - RDY: Data Ready Pin.
Python Integration	- I2C wiring instructions. - Library installation via Arduino IDE.
WipperSnapper	- Uses Adafruit CircuitPython library. - Wiring for CircuitPython boards and Raspberry Pi.
Calibration Methods	Enables IoT integration with Adafruit IO. Requires proper wiring, I2C scan, and configuration via the platform interface.
Applications	Environmental monitoring, HVAC systems, scientific research, and educational projects.

3.7 System Deployment

1. Hardware Configuration and Integration

The deployment begins by physically connecting the Adafruit SCD-30 sensor to the Raspberry Pi 5. The SCD-30 communicates over I2C, a standard protocol supported by the Raspberry Pi's GPIO pins.[11] This involves wiring the sensor's SDA and SCL lines to the corresponding pins on the Raspberry Pi, along with proper grounding and power (3.3V or 5V as required by the sensor's specifications). The Raspberry Pi 5, chosen for its robust processing capabilities, ensures smooth data acquisition and preprocessing before transmitting data to the storage layer.

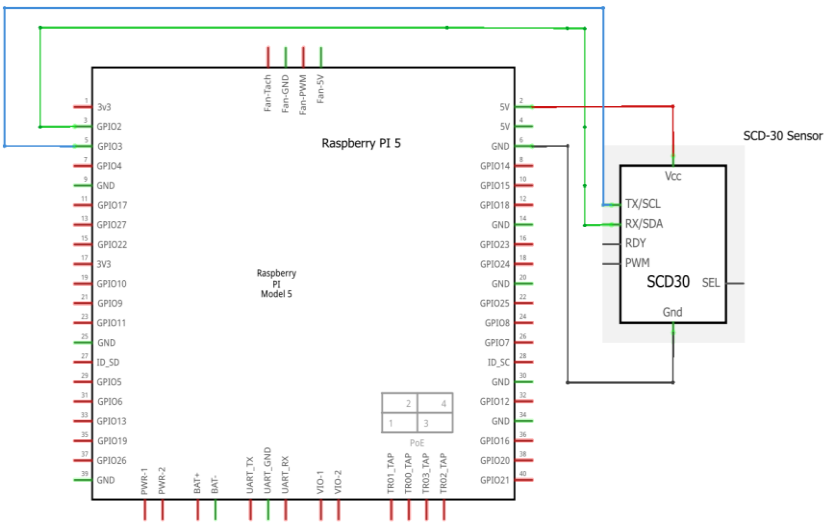

Figure 1 Raspberry Pi and Sensor circuit diagram

Table 8 SCD-30 Pinouts

SCD-30 Pin	Function	Raspberry Pi Pin	Purpose
VIN	Power (3.3V/5V)	Pin 1 (3.3V) or Pin 2 (5V)	Supplies power to the SCD-30 sensor from the Raspberry Pi. Ensures the sensor operates.
GND	Ground	Pin 6 (GND)	Connects the ground of the SCD-30 sensor to the Raspberry Pi for completing the circuit.
SDA	Data Line	Pin 3 (GPIO2/SDA)	Transmits data between the SCD-30 sensor and the Raspberry Pi via the I2C protocol.
SCL	Clock Line	Pin 5 (GPIO3/SCL)	Synchronizes communication between the SCD-30 sensor and the Raspberry Pi.

[11] *Adafruit SCD-30 Datasheet and Documentation https://www.adafruit.com/product/4867*

2. Software Environment Setup

To streamline code development and deployment, PlatformIO is utilized as an integrated development environment and build system. Although commonly associated with microcontroller platforms like Arduino or ESP32, PlatformIO can be configured to facilitate Python-based workflows on the Raspberry Pi. In this scenario, a Python script responsible for reading CO2 values from the SCD-30 is developed and version-controlled within a PlatformIO project structure. PlatformIO assists in managing dependencies, ensuring the correct libraries (e.g., Adafruit's CircuitPython SCD-30 library) and Python drivers are installed. As a result, the data collection script can be easily iterated upon, tested, and deployed to the Raspberry Pi.[12]

3. Data Ingestion and Storage with MongoDB

Once the sensor data is acquired, each reading—consisting of time stamped CO2 concentration (in ppm) is inserted into a MongoDB database. MongoDB offers a flexible schema-less model, allowing for the straightforward storage of time-series environmental data without rigid schema definitions.[12] By establishing a connection from the Raspberry Pi to a MongoDB instance in the cloud, incoming sensor data points are securely transmitted and recorded. This approach enables long-term historical data accumulation, forming the basis for model training and ensuring that the predictive analytics layer can operate on a rich dataset of past environmental conditions.[13]

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[12] *Adafruit SCD-30 Datasheet and Documentation https://www.adafruit.com/product/4867*
[13] *MongoDB for IoT https://www.mongodb.com/solutions/iot*

4. Model Training and Deployment

The predictive model, designed to forecast CO2 emissions or related environmental parameters, is trained using historical data retrieved from MongoDB. In practice, training can occur either directly on the Raspberry Pi—leveraging its 8GB of RAM and relatively powerful CPU—or more efficiently on a remote server. The training process involves the following steps:

Data Extraction and Preprocessing:

A separate Python script or Jupyter notebook connects to MongoDB to retrieve historical readings. It applies data cleaning methods, handles missing values, and performs feature engineering (e.g., rolling averages or time-based features) to prepare the dataset for training.[14] [15]

Model Training:

The forecasting model is trained on the prepared dataset Using libraries like scikit-learn and TensorFlow. This model learns patterns and trends from historical CO2 levels and other environmental variables, enabling it to predict future emissions.

Model Storage and Inference:

Once trained and validated, the model is serialized and stored (e.g., as a .pkl file). During inference, the Raspberry Pi retrieves recent sensor data from MongoDB, feeds it into the model, and obtains a prediction of future CO2 levels. This inference step can be automated, running at scheduled intervals or triggered on-demand.

[14] *Data Cleaning Techniques for Machine Learning* https://towardsdatascience.com/data-cleaning-techniques-for-machine-learning

[15] *feature Engineering for Time-Series Data* https://link.springer.com/chapter/10.1007/978-3-030-15729-6_5

5. System Operation and Maintenance

In a deployed environment, the system operates continuously:

The Raspberry Pi queries the SCD-30 sensor for new CO2 data at defined intervals and stores these readings in MongoDB.

Periodically, the model training pipeline is re-run (e.g., weekly or monthly) using updated data, ensuring the model remains accurate as operational conditions evolve.

Predictions obtained from the model inform decision-makers or automated control systems. For example, if predicted CO2 emissions exceed a certain threshold, the system could trigger alerts or prompt operational adjustments to improve sustainability outcomes.

6. Scalability and Extensibility

This system architecture can scale by adding more gases that can be detected utilizing other sensors and using MongoDB's horizontal scaling capabilities to handle increased data volume. Should the complexity of the model grow, or computation needs increase, the training process and inference services can be migrated to a more powerful cloud platform.

3.8 System Framework

The system framework for this project is proposed to support two different deployment approaches: a web-based application and an enterprise application, each for specific operational requirements. These frameworks provide flexibility, scalability, and functionality to address the varying needs of stakeholders in the oil industry.

Web-Based Application

The web application is lightweight and available for any user who needs a simple way to monitor emissions and plan for sustainability. The system provides a single interface where users can enter operational metrics, view emissions forecasts, and obtain actionable recommendations. Designed for accessibility, the application can be used on any device with internet connectivity, from desktops and tablets to smartphones.

It also provides core features such as real-time data visualization, thereby helping users track trends in emissions and metrics on performance via intuitive dashboards. The web application also provides integration through an API that can be seamlessly integrated with existing systems. This provides a cost-effective approach with minimal maintenance.

Enterprise Application

It targets large organizations with complex operations, thus providing a robust and scalable solution. This will integrate with internal systems, such as ERP, SCADA, and other proprietary systems that provide a complete view of the operation. Among the major features of the enterprise application, security is paramount, based on advanced protocols like data encryption and role-based access controls to protect sensitive information.

It also allows the customization of modules, through which organizations can customize their needs in forecasting algorithms, reporting tools, and compliance trackers. Designed for high performance, the enterprise application manages huge datasets and offers real-time insights for multiple operational units. This approach is particularly suitable for large oil corporations and multinational entities requiring detailed reporting and integration with enterprise-grade systems.

By providing these two deployment options, the system framework ensures that organizations can leverage predictive analytics to enhance sustainability in petroleum operations.

4 Detailed Project Requirement

4.1 Functional Requirements

CO_2 and Prediction

- **FR-1:** The system shall predict CO_2 emissions.
- **FR-2:** The system shall detect CO_2 emissions.
- **FR-3:** The system shall use oil refining output as a metric.
- **FR-4:** The system shall record CO_2 data.

Visual Representation

- **FR-5:** The system shall provide a visual representation of the outputs.
- **FR-6:** The system shall accurately detect CO2 levels in the environment using sensors.

User Requirements

- **FR-7:** The system shall accurately detect CO_2 levels in the environment using sensors.
- **FR-8:** the system shall have two different interfaces for regulators and oil companies.
- **FR-9:** the system shall be able to interface with the existing infrastructure.
- **FR-10:** the system shall allow environmental regulators to input test data for prediction.
- **FR-11:** the system shall authenticate the user based on his role (RBAC AUTH)

Integration and Reporting

- **FR-11:** The system shall provide APIs to enable integration with third-party systems for data exchange.
- **FR-12:** The system shall allow users to export reports in formats such as PDF or CSV.

4.2 Non-functional Requirements

Performance Requirements

- **NFR-1:** The system shall respond with fast model execution for decision-making.
- **NFR-2:** The system shall maintain CO_2 detection accuracy within a margin of error of +/- 5%.
- **NFR-3:** The system shall ensure prediction models have an accuracy rate of at least 90%, measured by appropriate metrics such as Mean Absolute Percentage Error (MAPE).

Scalability and Maintainability Requirements

- **NFR-4:** The system shall provide high interpretability and scalability of models.
- **NFR-5:** The system shall be scalable to support the addition of new sensors and geographic locations without affecting overall system performance.
- **NFR-6:** The system shall be designed in a modular fashion, allowing for updates to sensors, algorithms, or user interfaces without downtime.
- **NFR-7:** The system code shall be modular to facilitate easy updates.

Usability Requirements

- **NFR-8:** The system shall have a user-friendly interface with intuitive navigation.
- **NFR-7**: The system should have an uptime of 100%.
- **NFR-9:** The system shall offer role-specific workflows and dashboards for technical and non-technical users.

4.3 Software Platform Specification

Table 9 Software specifications

Software Name	Specification	Description
Python	Python 3.10 or above	It is a programming language used to build a ML model.
Jupyter Notebook	-	It's an open-source web tool for creating and sharing documents with live code, equations, visualizations, and text. It will be used to collaborate with the team in real-time and easily share the results.
Google Colab	-	Colab is a web-based Python editor that allows anyone to write and run Python code. It's notably useful for ML, data analysis
Platform.io	-	A framework utilizing C/C++ designed for developing applications using embedded systems, built on the VScode framework.
Docker	-	a set of platform as a service products that use OS-level virtualization to deliver software in packages called containers.
MongoBD	-	MongoDB is a source-available, cross-platform, document-oriented database program. Classified as a NoSQL database product

4.4 System Analysis

This part presents diagrams of the proposed system to provide a visual explanation of the system's functionality and how the system works.

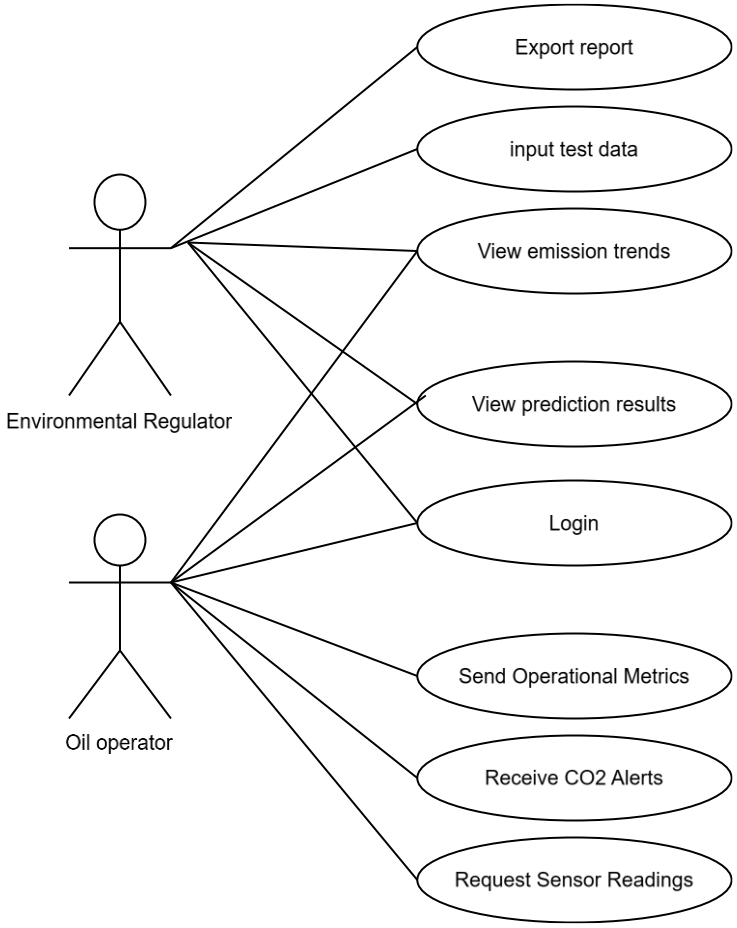


Figure 2 Use-Case Diagram

As illustrated in *Figure 2* Use-Case Diagram, the **Environmental Regulator** and **Oil Operator** interact with the system. Both can log in and create an account, with an error message displayed for login issues. The **Environmental Regulator** can export reports, input test data, view emission trends, and review predictions. The **Oil Operator** can send operational metrics, receive CO_2 alerts, and request sensor readings. While both can log in and view results, their roles differ: the **Regulator** focuses on emission data and predictions, and the **Operator** handles operational tasks and real-time metrics. The system supports both roles in CO_2 monitoring and oil production operations.

System Architecture

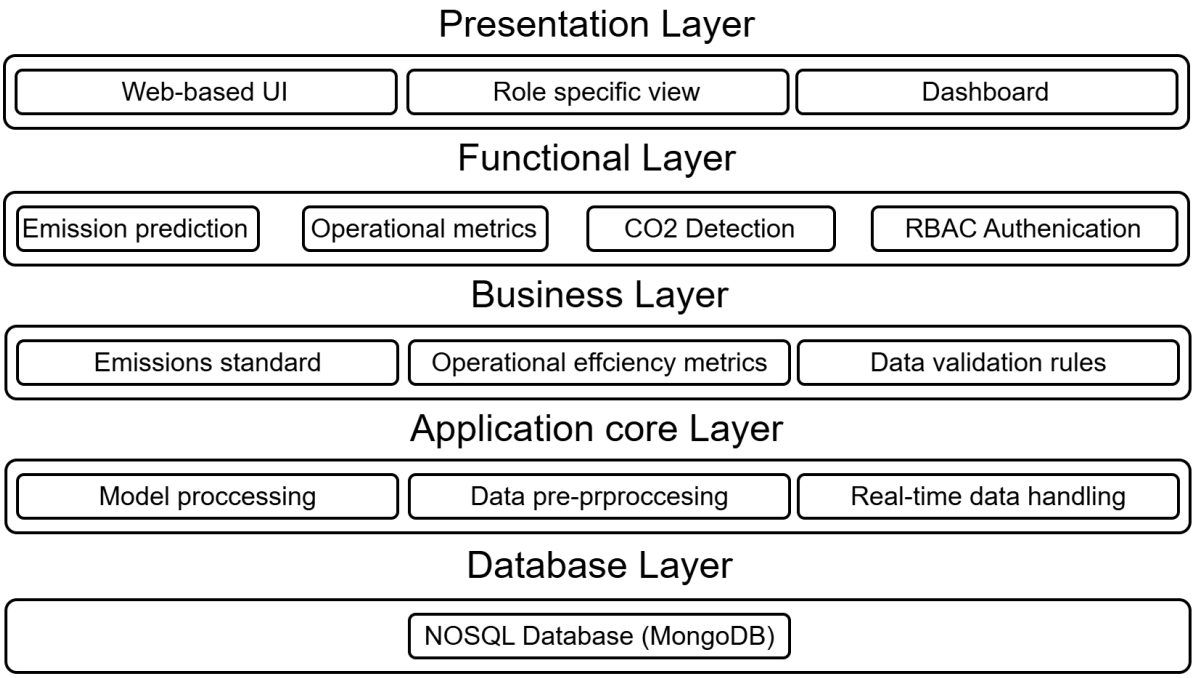


Figure 3 System layered Architecture

The diagram above figure 3 illustrates the system's layered architecture, consisting of four layers:

Presentation layer

The presentation layer is a layer that includes the components that the user is going to interact with, the presentation layer includes **Web-based UI** which is the main interface that the user will interact with. **A role specific view** will determine the kind of interface that will show to the user based on his role. The **Dashboard** component will show the available functions based on the role assigned.

Functional Layer

This layer will include the components that handle the main operations and logic. It ensures that the system performs tasks like calculations, validations, and decision-making. This layer includes **Emission prediction** which will predict the emissions based on the data that gets passed to it. **Operational metrics** are the products oof the refined oil that get produce during the refining process. **CO2 Detection** which will detect CO2 gas in real-time. **RBAC Authentication** which will give the role for a stakeholder based on their role in the sector.

Business Layer

This layer will include all the components that will be used to determine the outputs of functions based on the components that this layer holds. It includes **Emissions standard** which are legal regulations for emissions that are set in quantitative limits on every specific source over specific time frames. **Operational efficiency metrics** are key performance indicators that measure how effectively a system uses its resources. **Data validation rules** are a set of rules that will validate the data given.

Application core Layer

This layer will have the core process for the system. It includes **Model processing** which will have the prediction model ready to handle input data. **Data preprocessing** is the component that will preprocess the data that will be used by the model. **Real-time data handling** the component that handles the data given from the sensors.

Database Layer

This layer is the layer responsible for holding the database and is set at the end of the architecture to ensure data safety from any outside threat. This layer holds the database component **NOSQL Database (MongoDB)** which is responsible for holding data.

Activity Diagram

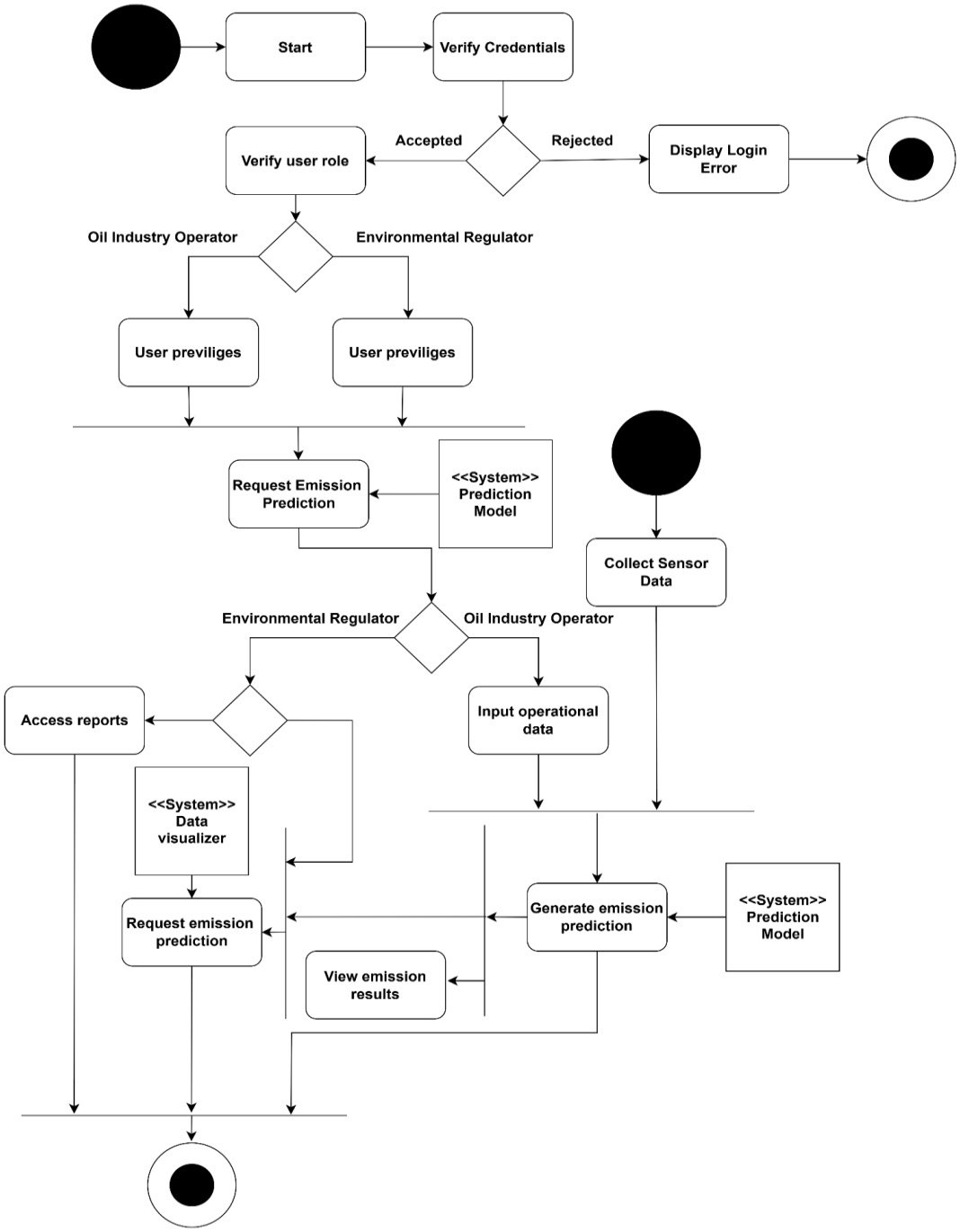


Figure 4 Activity Diagram

The activity model in figure 4, illustrates the system's dynamic activities and events for each user, highlighting their specific actions and roles.

Sequence Diagram

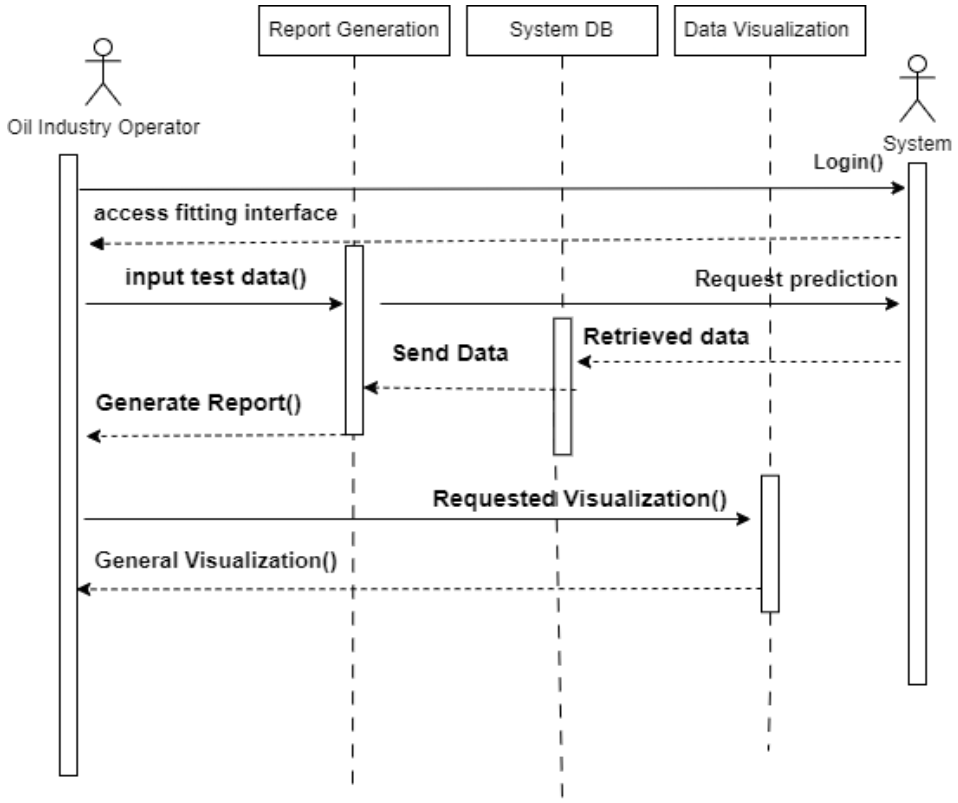


Figure 5 Oil Operator Action Sequence Diagram

Table 10 Oil Operator scenario

Data Input by Oil Industry Operator / Emissions Prediction Request / Data Visualization Request	
Actor	Oil industry Operator
Precondition	Oil industry Operator is logged in and has access to operational data input features.
Main Flow	1.Oil Industry Operator inputs operational data into the system. 2. The system requests emissions prediction based on the input data. 3. The prediction system returns the emissions prediction result. 4. The operator requests data visualization to view the emissions prediction and trends. 5. The system returns and displays the visualized data to the operator.
Postcondition	Emissions predictions are generated, and the data is visualized for the Oil Industry Operator to analyze.

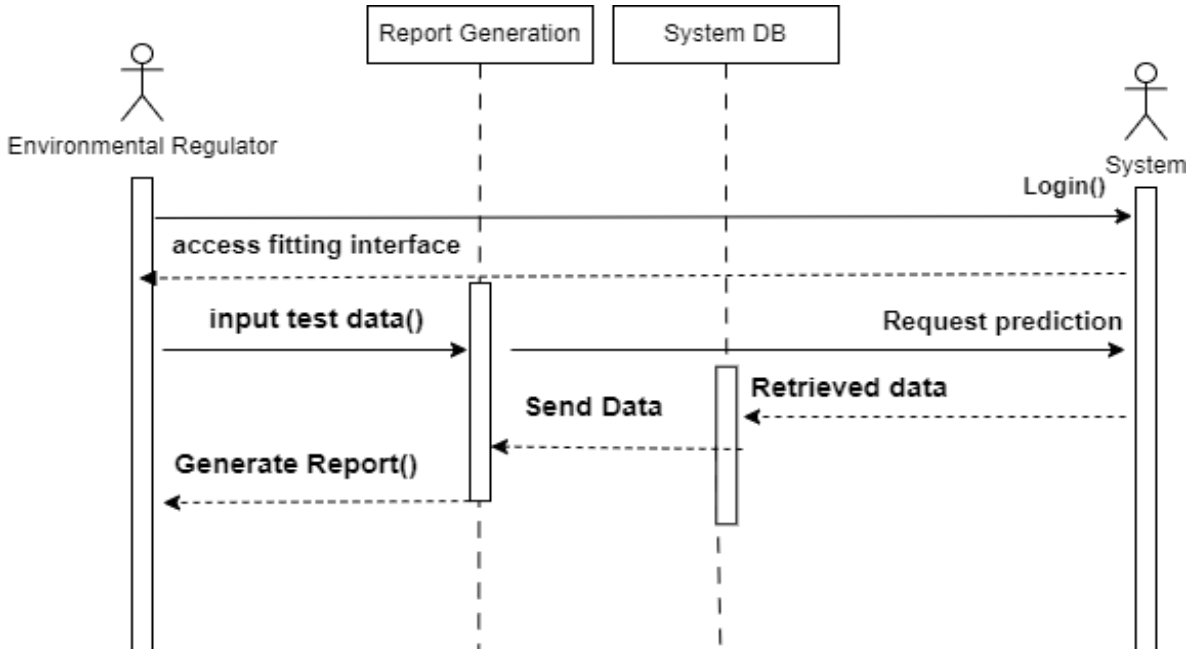


Figure 6 Regulator Action Sequence Diagram

Table 11 Environmental Regulator Scenario

Environmental Regulator Inputs Data	
Actor	Environmental Regulator
Precondition	The environmental regulator is logged in and has access to the reports feature.
Main Flow	1. The Environmental Regulator inputs test data into the system. 2. The system processes the test data and requests prediction results. 3. The system returns the prediction results based on the input test data.
Postcondition	The system generates predictions based on test data, and the results are displayed to the environmental regulator.

4.5 System Prototype

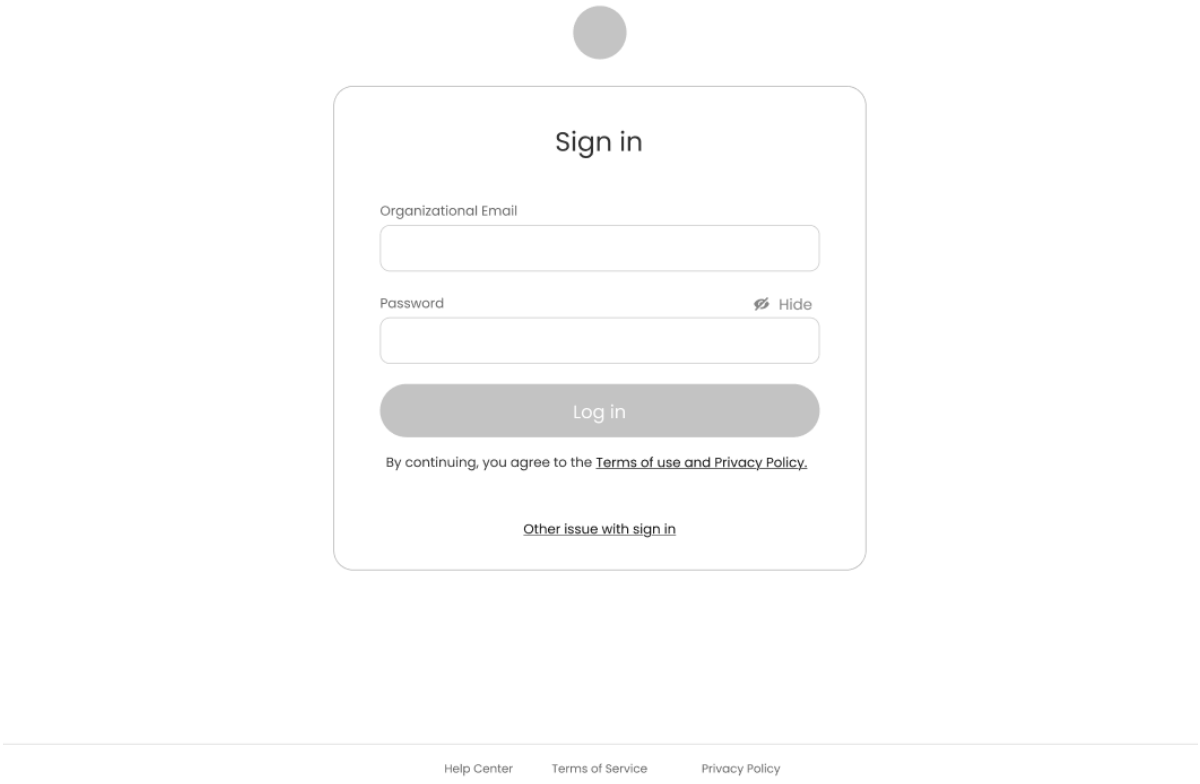


Figure 7 Proposed Sign-in Page

The system prototype includes a login page where users enter their Organizational Email and Password. Upon successful validation, the system grants access based on the user's role—Oil Industry Operator or Environmental Regulator. If the credentials are incorrect, an error message is displayed.

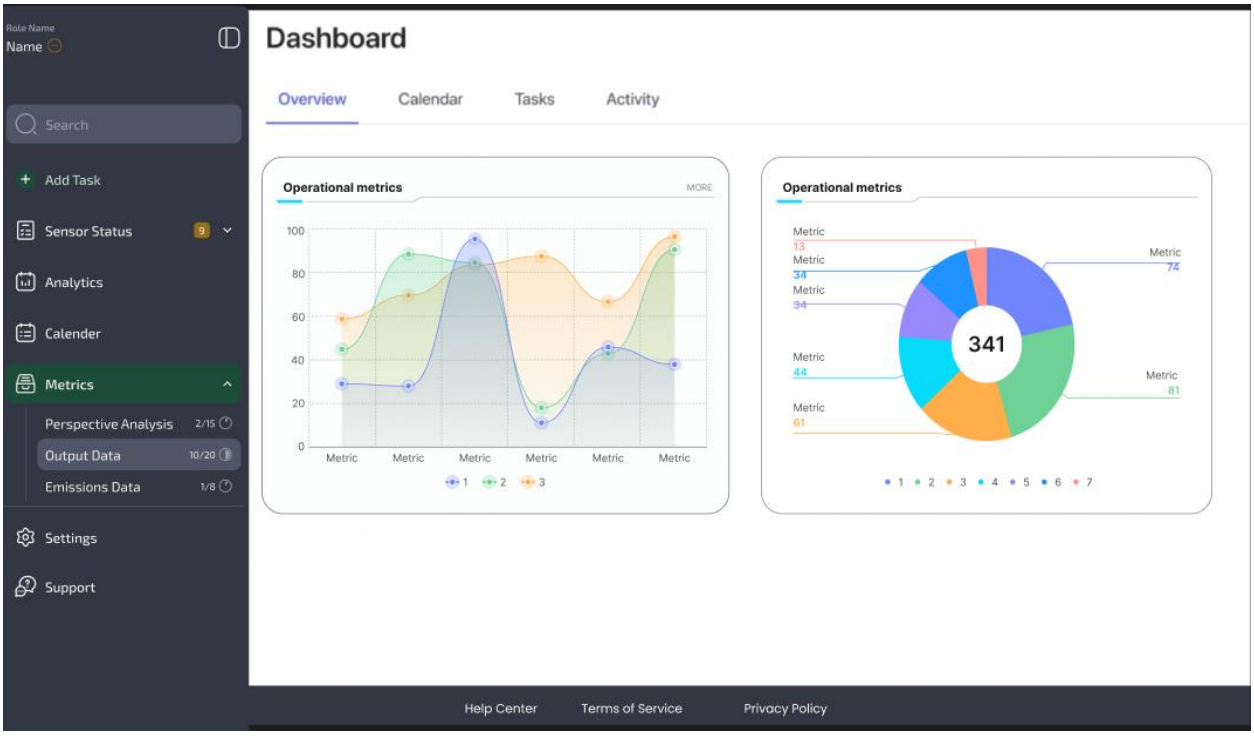


Figure 8 Proposed Oil Industry Dashboard Page

The dashboard for oil industry operators includes a vertical navigation menu on the left, providing easy access to various sections. Users can navigate to the "Add Task" section to create new tasks or operational activities, while the "Sensor Status" section displays the current status of sensors. The "Analytics" section offers access to detailed data insights, and the "Metrics" section, highlighted in green to show it is currently active, focuses on specific metric-related data. Additionally, there are subsections such as "Perspective Analysis," "Output Data," and "Emissions Data," enabling users to track progress within these categories effectively.

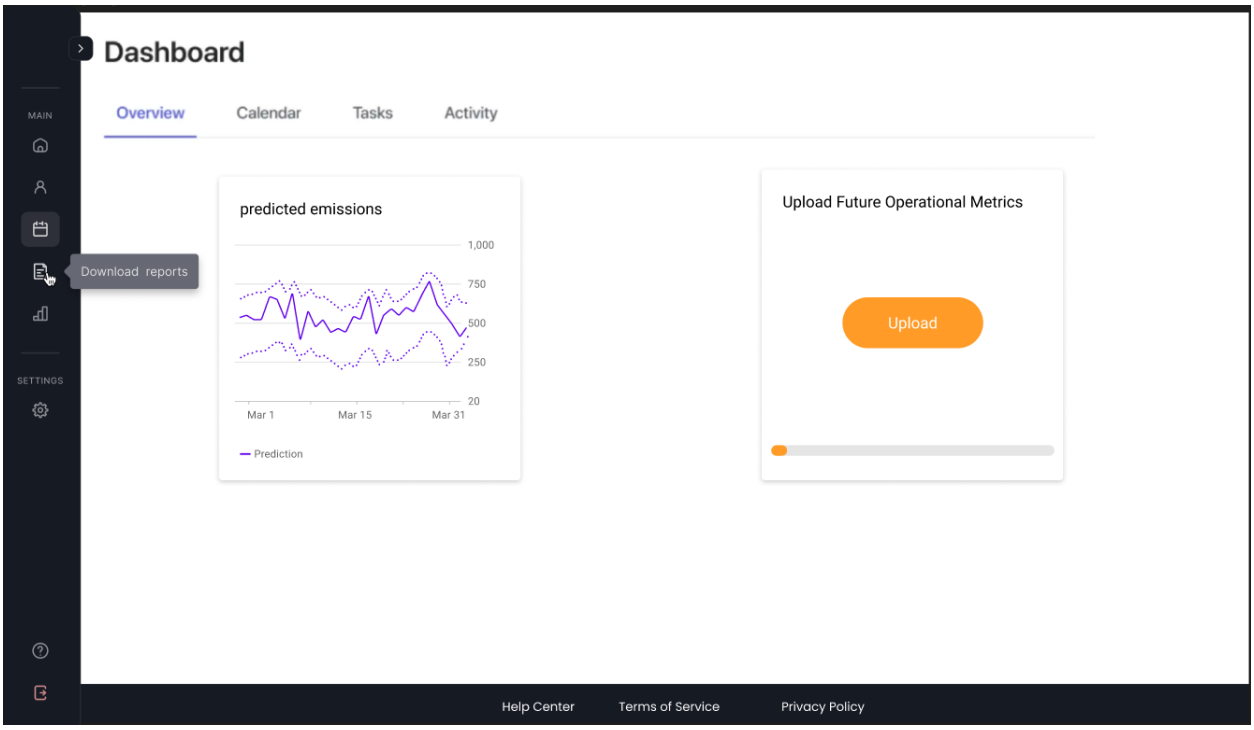


Figure 9 Proposed Environmental Regulators Dashboard Page

The dashboard designed for environmental regulators provides a user-friendly interface for monitoring emissions and managing data. Users have the flexibility to choose different visualization methods to analyze trends effectively based on their preferences or requirements. On the right, a dedicated "Upload Future Operational Metrics" section allows regulators to input data through an upload button, accompanied by a progress bar to indicate the upload status. A vertical navigation menu on the left offers quick access to key sections, including the main dashboard for an overview, analytics for detailed data insights, and settings for system configuration. This streamlined design ensures regulators can efficiently monitor emissions, input operational metrics, and manage environmental compliance with ease.

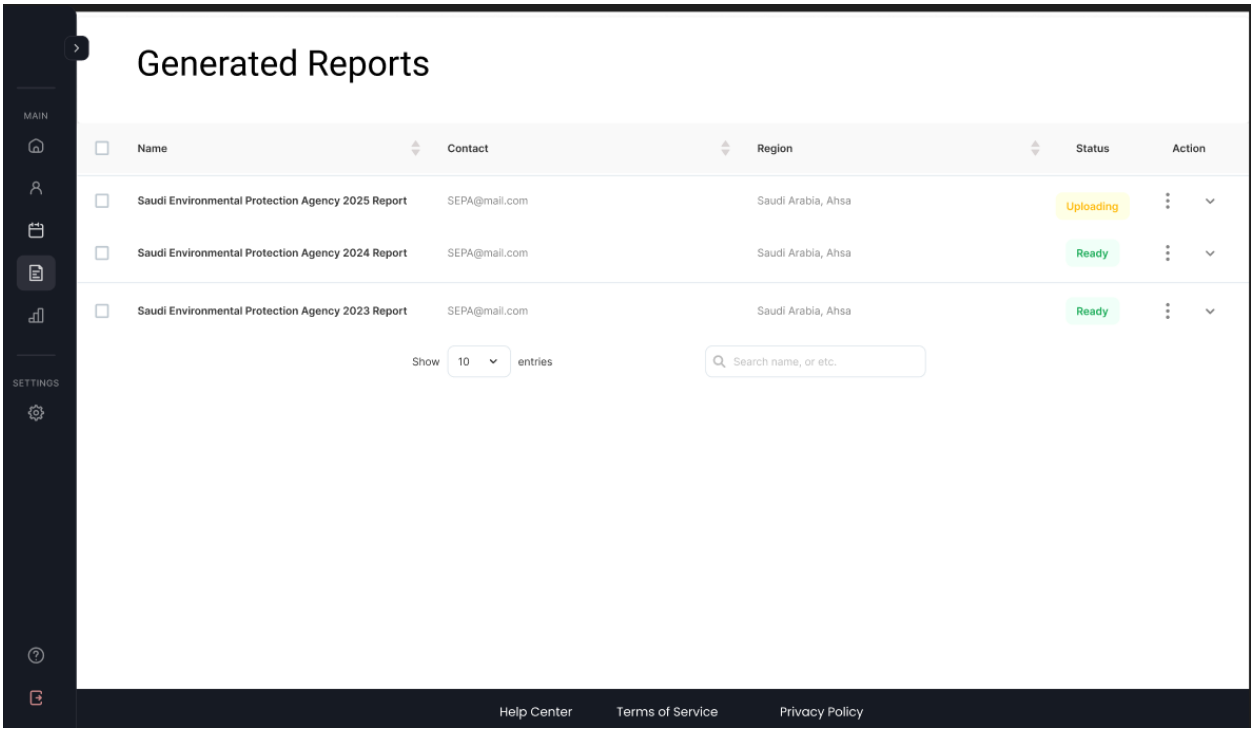


Figure 10 Proposed Report Generation and Access

The Reports Generation Page provides a centralized interface for managing generated reports through a well-organized tabular layout. The table includes columns for report name, associated contact information, region, status, and available actions. The status column uses color-coded indicators to represent the progress of each report, while the action menu allows users to manage reports individually. The page also features a search bar for filtering reports by various criteria and a dropdown menu to control the number of entries displayed per page.

5 Discussion of tools and techniques

This section discusses the tools and techniques used in the project proposal and those required for the project implementation.

5.1 Tools and Techniques Used in the Project Proposal:

- **Canva**: A web-based platform for drawing, revising, and sharing diagrams. It allows easy collaboration, making it ideal for creating system diagrams.
- **Figma**: A vector graphics editor and prototyping tool used for designing the system prototype, offering real-time collaboration.
- **Microsoft Office Word**: A word processing application used to document project reports.
- **Draw.io**: A free online diagram software for creating system analysis diagrams, ideal for designing flowcharts and other system representations.
- **Google Docs**: Used for document collaboration, allowing the team to access and edit the document simultaneously.

5.2 Tools and Techniques Required for Project Implementation:

- **Flask**: A web framework for building web applications. It will be used to integrate the machine learning model into the proposed web application. Also used for its NoSQL API Features.
- **XGBoost**: A powerful machine learning algorithm that will be used for classification and regression tasks in the project.
- **LSTM**: A deep learning model for sequential data that will be used for predicting time-series data.
- **Python**: A high-level programming language for building both the website and machine learning model.
- **Google Colab**: A cloud-based notebook for writing and executing Python code. It supports many machine learning libraries and facilitates collaboration.

- **Visual Studio**: An integrated development environment (IDE) used for developing computer programs, websites, and web apps, providing various development tools and compilers.
- **API Integration**: APIs will be developed for integrating the system with external sources and services.

Additional Tools for Project Implementation:

- **MongoDB**: A NoSQL database for storing operational data and sensor readings. MongoDB will be used to manage large volumes of unstructured or time-series data.
- **Docker**: For containerizing the application, ensuring portability and easy deployment across different environments.
- **Git**: A version control system used for tracking code changes and collaborating with the development team. **GitHub** will be used to manage the repository.
- **Jupyter Notebook**: For interactive development, testing, and visualizing machine learning models.
- **PyMongo**: The Python driver for MongoDB, used to interface with the database from the web application.
- **Matplotlib** and **Plotly**: Libraries for creating visualizations to help users monitor emissions trends, system performance, and operational metrics.

6 Details of project implementation conforming to the project proposal

This section describes the details of the current state of the system implementation. The implementation state is described based on four tasks. Hardware development, database development, machine learning development, and Front-end development.

6.1 Hardware development

The hardware development phase focused on setting up a live CO_2 sensing system using a Raspberry Pi 5 and an SCD-30 CO_2 sensor module. The objective was to collect real-time environmental CO_2 data at an interval of two seconds and 1 minute to transmit it for live monitoring and storage .

Hardware Setup

- **Device Used:** Raspberry Pi 5
- **Sensor:** Adafruit SCD-30 (CO_2, Temperature, Humidity sensor)
- **Connection Interface:** I2C communication between the Raspberry Pi and the SCD-30 sensor
- **Power Supply:** USB-C powered Raspberry Pi

Functionality

The SCD-30 sensor is configured to read only CO_2 values, excluding temperature and humidity data to reduce unnecessary processing. The sensor is programmed in Python using the **adafruit_scd30** and **adafruit-circuitpython-scd30** libraries. A custom script executes readings every interval and sends the CO_2 data to a remote MySQL database hosted on a docker container.

Communication

The Raspberry Pi transmits CO_2 values to the database using the **mysql-connector-python** library. The connection is made over the local network, targeting the IP address of the host machine (**192.168.100.xxx**) through port **3307** `which contains the docker container.` Robust exception handling is implemented to detect connection issues or sensor disconnection. If the sensor stops transmitting data, the live monitoring page automatically displays a system message indicating the loss of connection:
"No connection with the outer system. Last reading was more than 10 seconds ago."

Implementation Status

The hardware component is fully implemented and integrated with the backend system. The sensor is actively capturing CO_2 values, which are time stamped and reliably inserted into the database every interval.

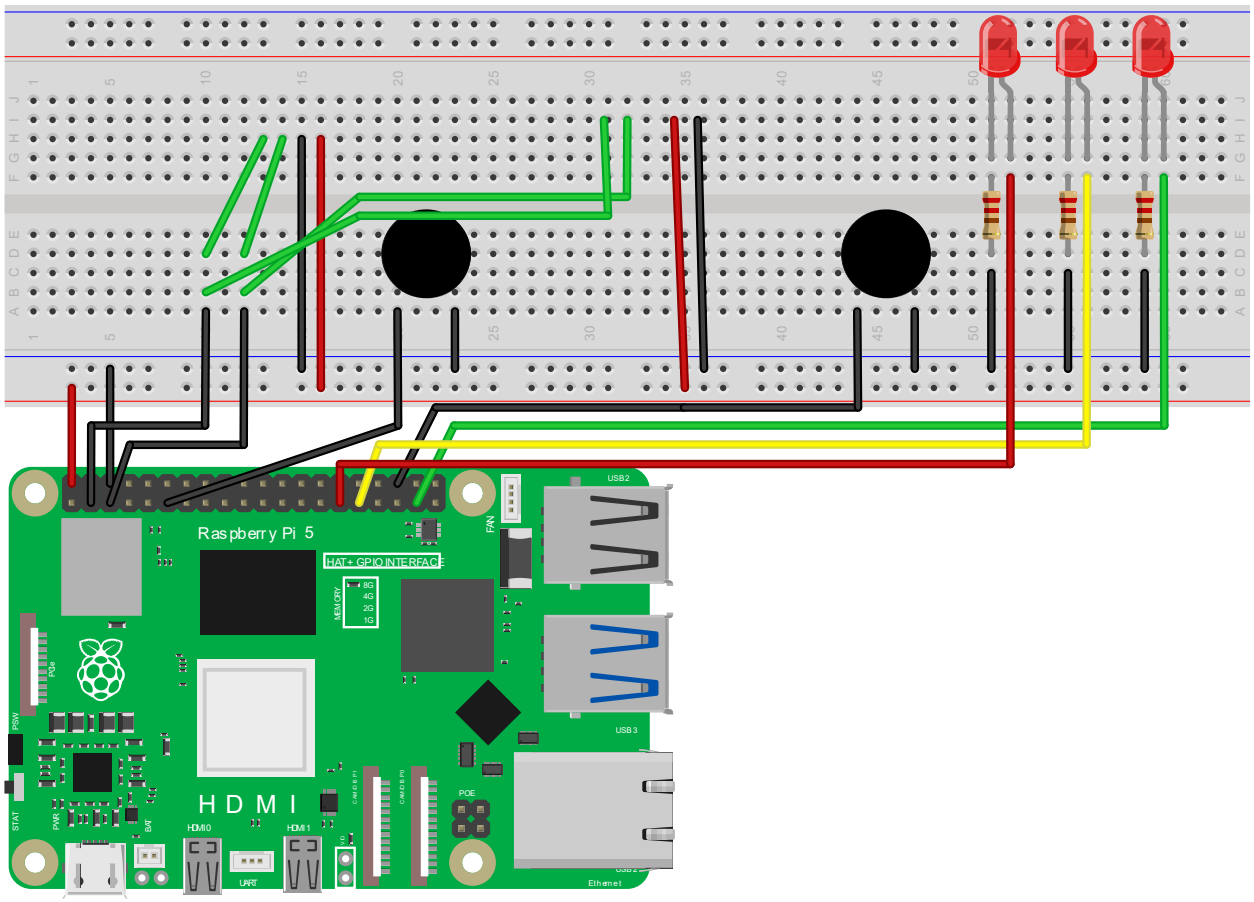


Figure 11 Implemented Circuit Diagram

6.2 Database development

The database development phase centered on designing and implementing a structured relational database to store, retrieve, and summarize CO_2 readings in real time. A MySQL database was selected due to its reliability, ease of use, and compatibility with both the Raspberry Pi and the Flask web framework.

Database Structure

The database schema was designed to reflect the three core functions of the system: real-time CO_2 recording, periodic summarization, and operator fuel input tracking. Accordingly, three distinct tables were created:

- **co2_readings:** Stores CO_2 values collected by the Raspberry Pi every 2 seconds. Each entry includes the measured CO_2 value and a timestamp.
- **co2_summary:** Contains 1-minute statistical summaries (MIN, MAX, AVG) of the data recorded in **co2_readings**, along with the timestamp of the summary interval.
- **operator_inputs:** Records data entered by oil operators, including the amounts of crude oil, diesel, gasoline, and kerosene. Each record includes the operator ID, operator name, input values, and the time of submission.

The **create_tables.py** script was used to initialize and configure these tables within the database.

Docker-Based Deployment

To ensure consistency and portability, the MySQL database was containerized using Docker. A **docker-compose.yml** file was created to define and manage the database service, including volume mounting for persistent storage and custom port binding (port **3307**) to avoid conflicts with other local services. This setup allows the database to be deployed and maintained independently of the host system, improving modularity and simplifying updates.

This created and started the MySQL container, ensuring the database was isolated from the host environment but fully accessible by the Raspberry Pi and the Flask application over the local network.

Real-Time and Summary Processing

Data insertion into **co2_readings** is performed by the Raspberry Pi through the mysql-connector-python library. A separate background thread running within the Flask backend is responsible for generating 1-minute summaries by querying the most recent readings from **co2_readings**, calculating statistics, and inserting results into **co2_summary.**

The summary generator also includes exception handling to ensure that no summaries are produced when the sensor is inactive. In this case, the system logs a warning internally and refrains from updating the database with inaccurate or stale data.

Implementation Status

The database is fully functional and integrated with both the sensor hardware and the web application. Real-time readings, summaries, and operator inputs are reliably stored and retrieved. The use of Docker has enhanced the maintainability and reproducibility of the database environment.

6.3 Machine Learning Development

Dataset Description and Preparation

The primary focus is on **carbon dioxide (CO_2)**, with supplementary GHG features (methane, nitrous oxide, oil_CO_2, gas_CO_2) included to enhance the model's predictive performance.

Preprocessing and Merging

The SAMA dataset underwent extensive cleaning:

- Non-numeric rows were filtered.
- Column headers were renamed for consistency.
- Missing values were imputed using mean substitution.

Following preprocessing, the datasets were merged on the **year** column using an **inner join**, ensuring consistency across both emissions and fuel production metrics.

Feature Engineering

A new feature, **cumulative_ghg**, was engineered by summing the cumulative historical values of CO_2, methane, and nitrous oxide.

This feature effectively represents the ongoing climate burden and demonstrates a high correlation with annual CO_2 emissions.

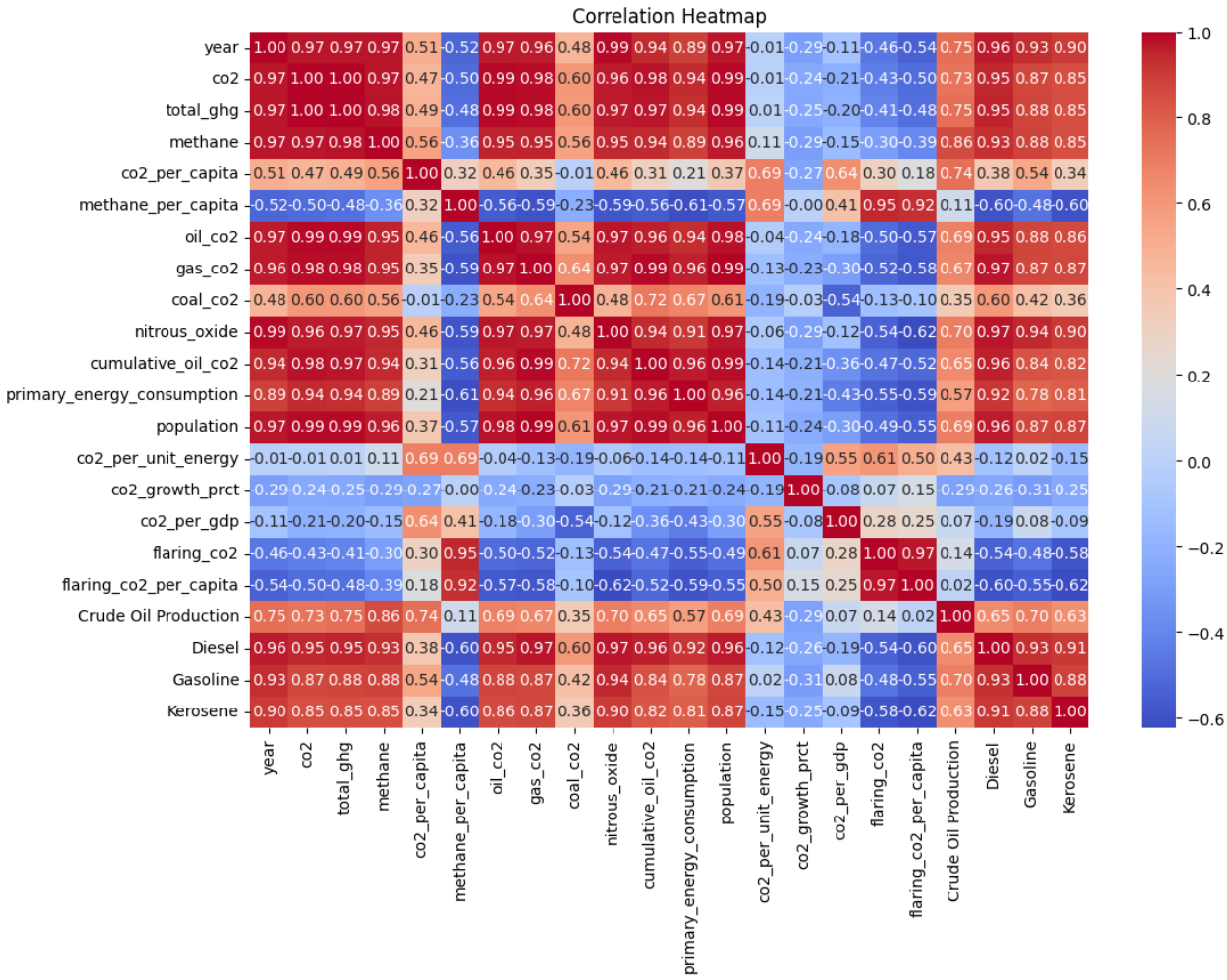


Figure 12 Merged Datasets Correlation heatmap

The correlation heatmap reveals strong positive relationships between CO_2 and features like **total_ghg** and **oil_co2**, indicating that these variables closely align with CO_2 emission trends. This correlation supports their use in the model as predictors or complementary outputs, enhancing its accuracy and interpretability. By focusing on statistically aligned variables, the model's architecture remains both data-driven and grounded in real-world emission dynamics.

Feature Importance and Selection

Feature selection was guided using ANOVA F-score (via **SelectKBest**) and correlation analysis. Key predictors included:

- **total_ghg**
- **oil_CO_2**
- **gas_CO_2**
- **diesel**
- **cumulative_ghg**

Table 12 Feature importance

Rank	Feature	F-score	P-score
1	total_ghg	49880.824008	8.003850e-87
2	methane	1079.667655	3.501819e-39
3	co2_per_capita	16.393747	1.545021e-04
4	methane_per_capita	19.619282	4.246048e-05
5	oil_co2	3862.866317	8.970457e-55
6	gas_co2	1753.073871	4.835552e-45
7	coal_co2	33.123325	3.438510e-07
8	nitrous_oxide	754.863585	6.064204e-35
9	cumulative_oil_co2	1139.763306	7.859908e-40
10	primary_energy_consumption	451.184940	4.827361e-29
11	population	2535.892095	1.438487e-49
12	co2_per_unit_energy	0.008510	9.268170e-01
13	co2_growth_prct	3.668130	6.039755e-02
14	co2_per_gdp	2.770285	1.014230e-01
15	flaring_co2	12.942447	6.651747e-04
16	flaring_co2_per_capita	18.963200	5.493261e-05
17	Crude Oil Production	67.481869	2.676669e-11
18	Diesel	527.386230	8.390691e-31
19	Gasoline	181.428725	1.658077e-19
20	Kerosene	148.459319	1.248517e-17

LSTM Model Development

Motivation and Setup

The primary motivation behind using a Long Short-Term Memory (LSTM) model was its capacity to learn complex temporal dependencies across multiple time steps. In the context of greenhouse gas (GHG) emissions forecasting, emissions are inherently sequential and influenced by both recent and long-term trends in fuel consumption and production behaviors. Traditional machine learning models may not capture these intricate patterns as effectively as sequence-based architectures.

Given the temporal structure of our dataset—spanning several years of fuel consumption and emission data—we hypothesized that leveraging an LSTM model would enable more accurate forecasting, especially when dealing with delayed environmental responses or cumulative effects such as CO_2 accumulation.

Feature Engineering and Input Design

To prepare the dataset for LSTM modeling, we performed the following:

Selected Input Features:

- Diesel
- Gasoline
- Kerosene
- Crude Oil Production
- CO_2 Emissions from the Previous Year (**oil_CO_2**)
- Cumulative Greenhouse Gas Emissions (**`cumulative_ghg`**)

Model Architecture and Training Strategy

The final architecture adopted was relatively lightweight but expressive enough for the task:

Layer Configuration:

- **`LSTM Layer (64 units, return_sequences=True)`** – captures sequential patterns across full input sequences.
- **`Dropout (0.2)`** – mitigates overfitting by randomly deactivating neurons during training.
- **`LSTM Layer (32 units)`** – reduces dimensionality and focuses on compressed temporal representations.
- **`Dense Output Layer (6 neurons)`** – produces predictions for six target variables.

Training Parameters:

- **Loss Function**: Huber Loss, chosen for its robustness to outliers.
- **Optimizer**: Adam, known for fast convergence and adaptability.
- **Validation Strategy**: Leave-One-Out Cross-Validation (LOOCV) was employed to maximize training data utilization while fairly evaluating model generalizability, especially beneficial for smaller time-series datasets.

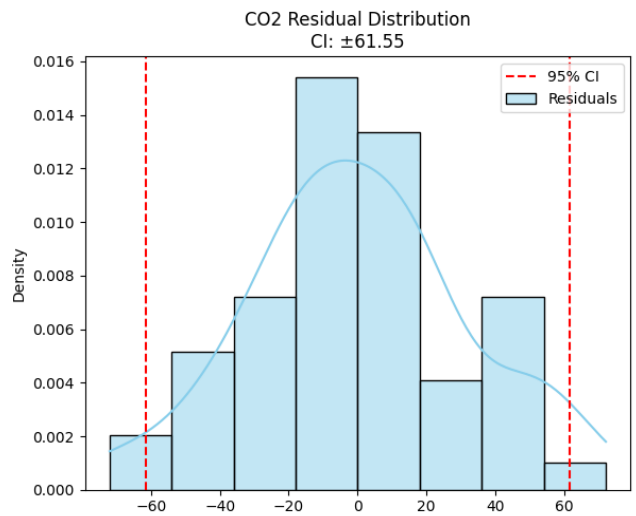


Figure 13 LSTM Confidence Interval

This figure shows LSTM CO_2 Prediction with Confidence Interval. The confidence interval (±61.55) is relatively wide, indicating that while the model captures the trend, there is some uncertainty in its point predictions. The presence of a few extreme residuals hints at possible underfitting in some cases or sensitivity to outliers.

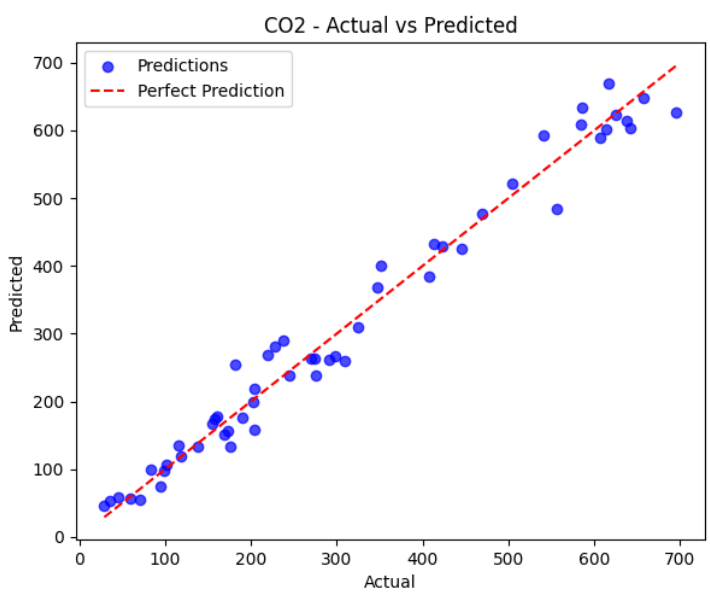


Figure 14 LSTM Residual Plot

This figure shows how close the LSTM model's predictions match actual values for each target. The closer the blue dots are to the red diagonal line, the better the model's accuracy. For CO_2, methane, and nitrous oxide, the predictions closely follow the actual values, indicating fairly strong model performance, however the model seems to struggle with CO_2 as it gets larger. Overall, the model performs best on the main greenhouse gases.

Performance Summary

While the LSTM model captured key trends and showed high R^2 for several targets, it underperformed slightly on CO_2 compared to XGBoost. Performance metrics for CO_2 were:

Table 13 LSTM Performance Summary

Metric	R^2	MAE	MSE	MAPE	sMAPE
Value	0.96	27.5	1449	4.3%	4.4%

Despite promising results, training time and CI width indicated reduced generalization capacity.

XGBoost Model Development

Motivation

We adopted XGBoost, a gradient boosting framework highly optimized for structured, tabular data. Its ability to handle complex feature interactions, outliers, and missing values made it well-suited for our greenhouse gas emissions dataset. XGBoost also offers significantly faster training times and more stable performance under Leave-One-Out Cross-Validation (LOOCV), which was critical given the relatively small number of yearly records. Although our primary objective was to predict CO_2 emissions, we included additional greenhouse gases—such as methane and nitrous oxide—as output targets to exploit potential correlations and improve CO_2 prediction accuracy through multi-output learning.

Implementation Strategy

We adopted a structured and modular machine learning pipeline using **Scikit-learn**, designed to maximize preprocessing robustness and reproducibility. The pipeline consisted of:

- **Imputation**: Handling missing data using **`SimpleImputer`** with mean strategy.
- **Feature Scaling**: Applied **`StandardScaler`** to ensure that XGBoost's gradient descent steps are balanced across features.
- **Model Wrapping**: Used **`MultiOutputRegressor`** to allow XGBoost to predict multiple GHG outputs simultaneously.
- **Regressor Core**: Each target was modeled independently using a **`XGBRegressor`** instance..

Hyperparameter Tuning with LOOCV

To ensure model generalizability and avoid overfitting, we applied **Leave-One-Out Cross-Validation (LOOCV)** during hyperparameter tuning using **`GridSearchCV`**. This was computationally intensive but ideal for our relatively small dataset. The parameter grid explored combinations of:

- **`n_estimators`** (100, 200): Controls the number of boosting rounds.
- **`max_depth`** (3, 5): Manages tree complexity and feature interactions.
- **`learning_rate`** (0.01, 0.05, 0.1): Balances convergence speed with stability.
- **`subsample`** (0.8, 1.0): Helps regularize by training on random row subsets.
- **`colsample_bytree`** (0.8, 1.0): Prevents overfitting by sampling features per tree.

The best hyperparameter configuration was selected based on **minimizing the negative mean squared error** across LOOCV folds.

Evaluation Results

After fitting the optimized pipeline, we evaluated its performance across six greenhouse gas targets. The XGBoost model exhibited consistently strong results, with:

- **High R^2 values**, reflecting excellent goodness-of-fit
- **Low MAE and MSE**, confirming accurate point predictions
- **MAPE and sMAPE metrics** under 5% for most targets, signifying excellent percentage-based error performance

These metrics not only demonstrated the model's **accuracy**, but also its **consistency**, especially for **CO_2**, **methane**, and **nitrous oxide**—the primary greenhouse gases of concern.

Advantages Over Sequential Models:

Compared to LSTM:

- XGBoost provided **faster training** and **simpler deployment**
- It required **less data engineering** (no sequence generation)
- Delivered **tighter confidence intervals** and **lower error margins** on average, especially under LOOCV

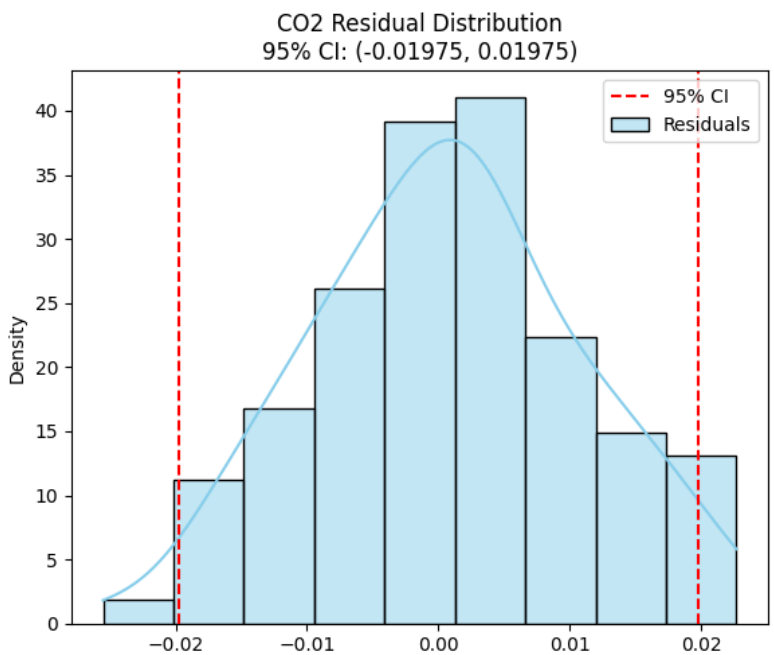


Figure 15 XGBoost Confidence Interval

This figure displays XGBoost residual plots. It reveals errors centered around zero with narrow, symmetric 95% confidence intervals, suggesting unbiased and stable predictions. The tight distribution of CO2 residuals reflects the model's high accuracy and low variance. This supports its robustness for deployment and interpretable forecasting.

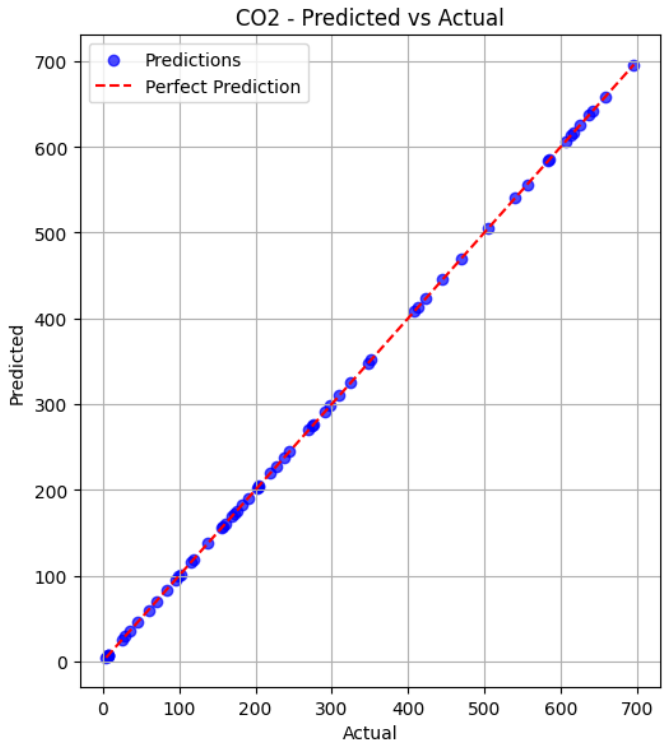


Figure 16 XGBoost Residual Plot

This figure displays XGBoost Predicted vs Actual Plots. These plots confirm strong agreement between predicted and actual values for all outputs, especially **CO2**. Predictions closely follow the diagonal, indicating excellent accuracy and minimal overfitting. Slight deviations in secondary outputs do not detract from the model's overall performance.

Results

XGBoost achieved **nearly perfect predictions for CO_2** and all other outputs. Confidence intervals were tight, and residuals were normally distributed and centered around zero.

Table 14 XGBoost Performance Summary

Metric	R^2	MAE	MSE	MAPE	sMAPE
Value	~0.98	0.0374	0.0023	0.074%	0.074%

Limitations of Ensemble Learning Between Current Models

While ensemble learning is a powerful strategy for improving model performance by combining multiple predictive models, it is not always straightforward, especially when the underlying models differ significantly in structure, training data, and preprocessing. In our case, several critical factors prevent the immediate application of ensemble learning between our developed LSTM and XGBoost models:

1. Different Data Structures and Input Paradigms

- The **LSTM model** processes **temporal sequences** using a sliding window of six years, learning from the order and trends of emissions over time.
- The **XGBoost model**, however, treats each year as an **independent tabular observation**, with no awareness of temporal context.

This fundamental difference in input structure makes it difficult to synchronize predictions from both models or merge them meaningfully in an ensemble framework.

2. Incompatible Preprocessing Pipelines

- LSTM uses **MinMaxScaler** across the full feature-target space for sequential data.
- XGBoost uses **SimpleImputer** followed by **StandardScaler** on flattened rows.

This leads to scaling mismatches that affect prediction comparability, requiring careful inverse transformation before any ensemble, which adds complexity and risk of distortion.

3. Uneven Performance and Generalization

Performance evaluation revealed key behavioral differences:

- **XGBoost** produced **consistently high R^2 scores (e.g., ~0.98)**, with narrow prediction intervals and low MAE/MAPE values. Its predictions were stable and generalized well under LOOCV.
- **LSTM**, while effective at capturing trends, showed signs of **underfitting** on CO_2 in high-emission years and **occasional overfitting** during LOOCV due to sensitivity to outliers in sequence data. This was also reflected in **wider confidence intervals (±61.55)** and **higher variance** in residuals.

  Given this discrepancy, averaging or blending outputs from both models could lead to **diluted performance**, especially where one model (e.g., LSTM) consistently deviates from ground truth in edge cases.

6.4 Front-end Development

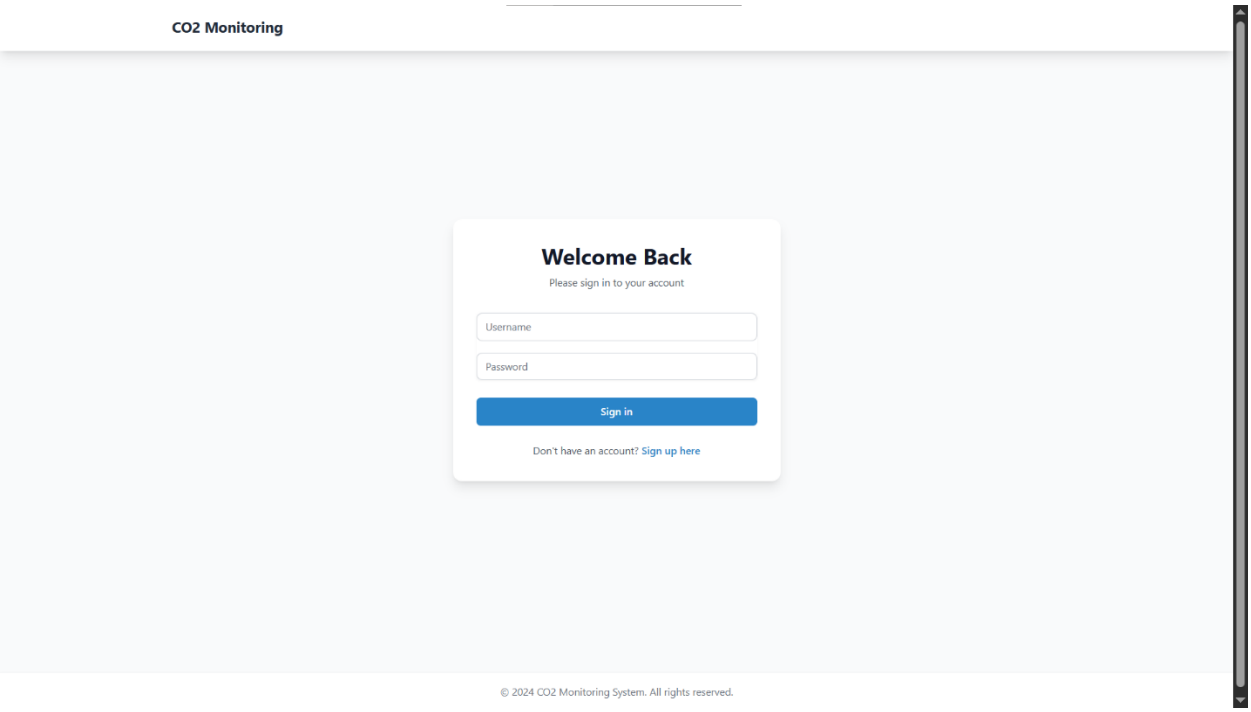


Figure 17 Implemented Sign In Page

This is the gateway to the system for both **operators** and **regulators**. The page provides fields for entering a username and password. Upon successful login, the system checks the role of the user and redirects them accordingly—operators are taken to the operational metrics input form, while regulators are redirected to the emissions prediction panel. A link at the bottom guides users to the sign-up page if they do not have an account.

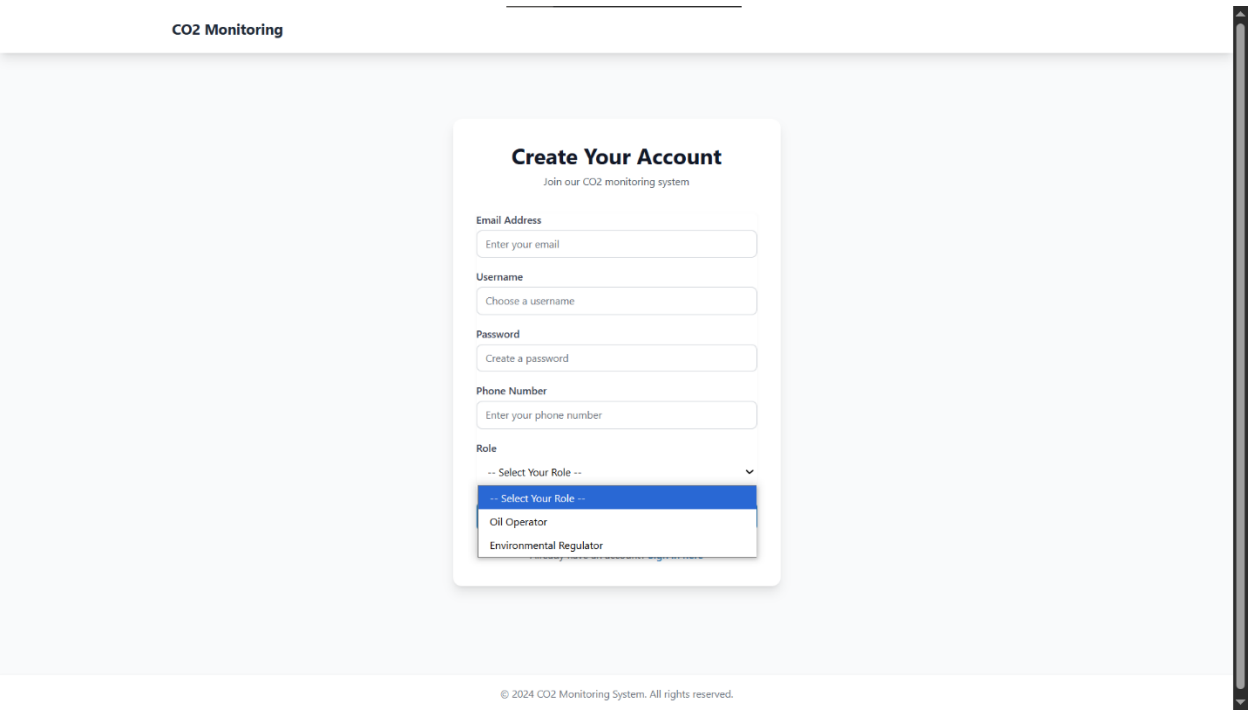


Figure 18 Implemented Sign Up Page

New users can create an account by providing their email, username, password, phone number, and selecting their role. The form is well-structured and visually in line with the login page, creating a cohesive authentication experience. The role selection here determines how the user will be routed after login, enforcing proper access control across the system.

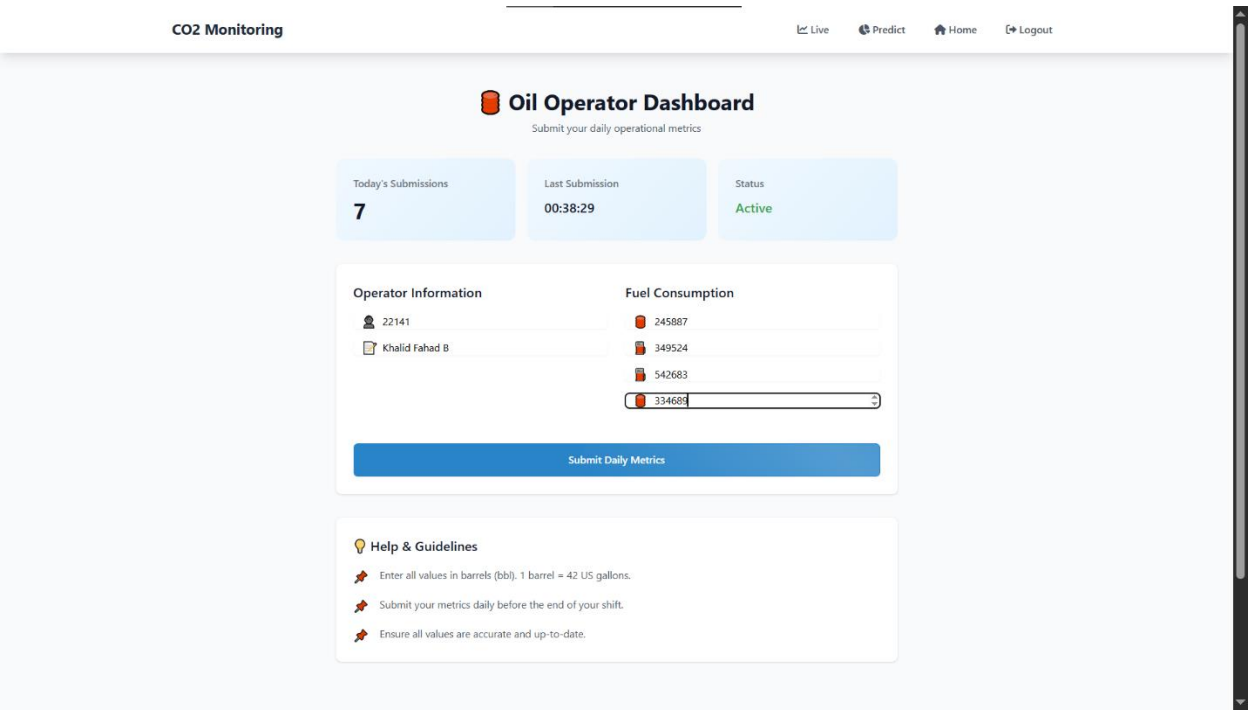


Figure 19 Implemented Oil Operator Dashboard

This page is designed specifically for users with the "Operator" role. It presents a structured and easy-to-follow form where the operator submits real-time data such as the quantity of Crude Oil, Diesel, Gasoline, and Kerosene. The interface includes placeholder examples for each input to help reduce user error. After submitting, this data is stored and used later for emission predictions. Navigation buttons at the top give operators access to other parts of the system, like live sensor monitoring and emission prediction, along with a visible Sign Out button for security.

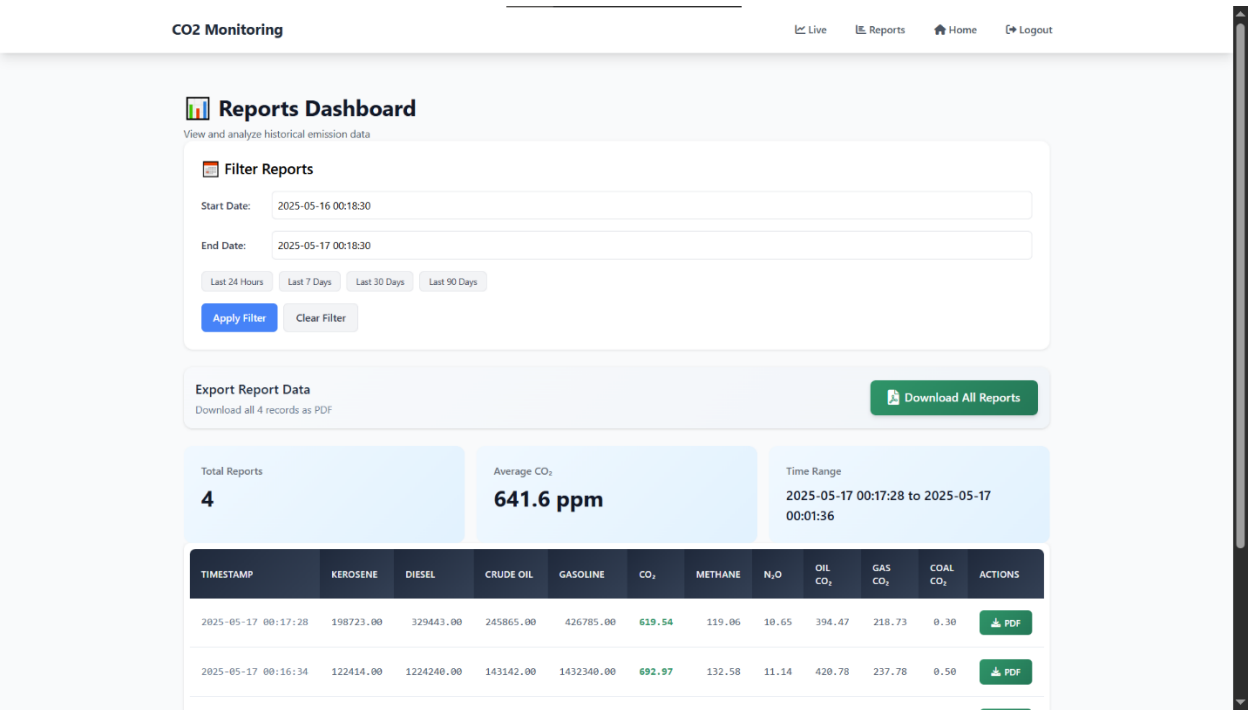
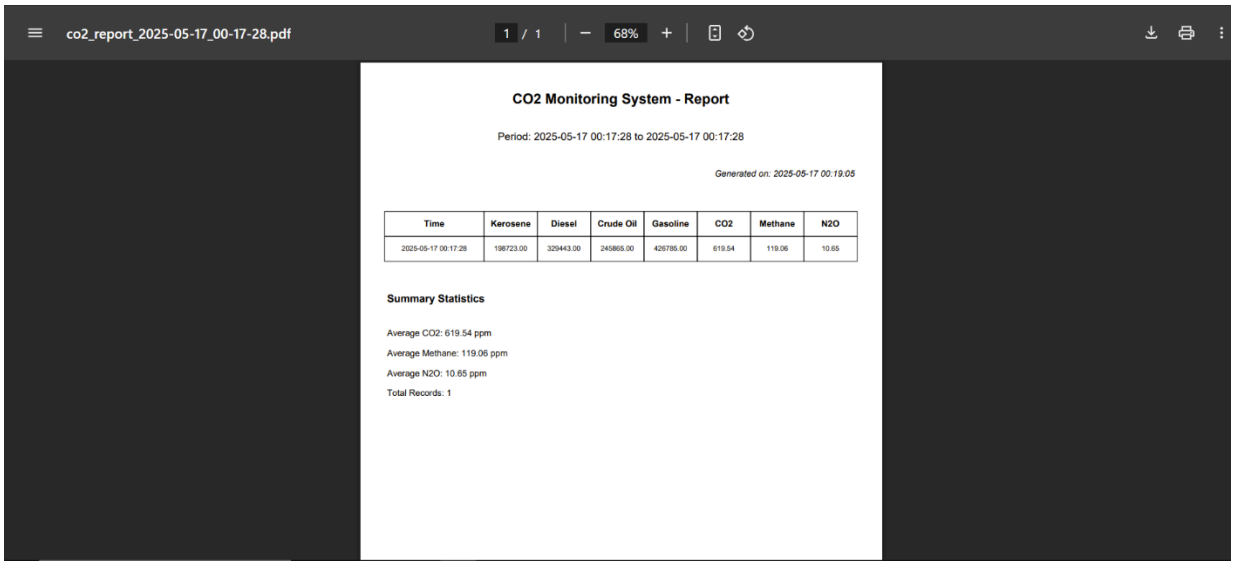


Figure 20 implemented Reports Log Page

Regulators use this dashboard to access all previous reports stored in the database. Each entry displays input data and its associated prediction results, along with timestamps. The layout includes a date-time filter for narrowing down specific reports and a pagination system to handle long lists. It's a vital part of the system that supports compliance checks and long-term tracking. The design remains clean and responsive with navigation and sign-out clearly accessible.



co2_report_2025-05-17_00-17-28.pdf

1 / 1 68%

CO2 Monitoring System - Report

Period: 2025-05-17 00:17:28 to 2025-05-17 00:17:28

Generated on: 2025-05-17 00:19:05

Time	Kerosene	Diesel	Crude Oil	Gasoline	CO2	Methane	N2O
2025-05-17 00:17:28	198723.00	329443.00	245865.00	426785.00	619.54	119.06	10.65

Summary Statistics

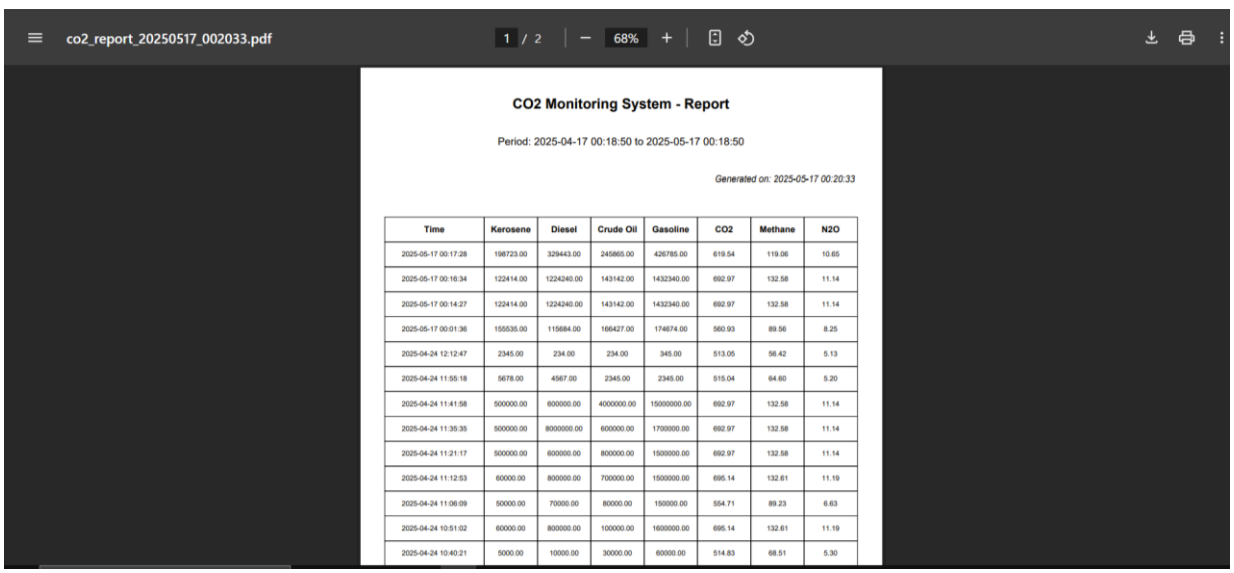
Average CO2: 619.54 ppm

Average Methane: 119.06 ppm

Average N2O: 10.65 ppm

Total Records: 1

Figure 21 Implemented Single Report Download



co2_report_20250517_002033.pdf

1 / 2 68%

CO2 Monitoring System - Report

Period: 2025-04-17 00:18:50 to 2025-05-17 00:18:50

Generated on: 2025-05-17 00:20:33

Time	Kerosene	Diesel	Crude Oil	Gasoline	CO2	Methane	N2O
2025-05-17 00:17:28	198723.00	329443.00	245865.00	426785.00	619.54	119.06	10.65
2025-05-17 00:16:34	122414.00	1224240.00	143142.00	1432340.00	692.97	132.58	11.14
2025-05-17 00:14:27	122414.00	1224240.00	143142.00	1432340.00	692.97	132.58	11.14
2025-05-17 00:01:36	155535.00	115684.00	166427.00	174674.00	560.93	89.56	8.25
2025-04-24 12:12:47	2345.00	234.00	234.00	345.00	513.05	56.42	5.13
2025-04-24 11:55:18	5678.00	4567.00	2345.00	2345.00	515.04	64.60	5.20
2025-04-24 11:41:58	500000.00	600000.00	4000000.00	15000000.00	692.97	132.58	11.14
2025-04-24 11:35:35	500000.00	8000000.00	600000.00	1700000.00	692.97	132.58	11.14
2025-04-24 11:21:17	500000.00	600000.00	800000.00	1500000.00	692.97	132.58	11.14
2025-04-24 11:12:53	60000.00	800000.00	700000.00	1500000.00	695.14	132.61	11.19
2025-04-24 11:06:09	50000.00	70000.00	80000.00	150000.00	554.71	89.23	6.63
2025-04-24 10:51:02	60000.00	800000.00	100000.00	1600000.00	695.14	132.61	11.19
2025-04-24 10:40:21	5000.00	10000.00	30000.00	60000.00	514.83	68.51	5.30

Figure 22 Implemented Multi reports download

The system provides users with the ability to export CO_2 monitoring data as PDF reports. Users can choose to download a single report for a specific record or export all available records within a selected time range as a comprehensive PDF file. Each PDF report includes detailed metrics such as fuel usage, greenhouse gas emissions, and summary statistics, enabling easy sharing, documentation, and offline analysis of environmental data.

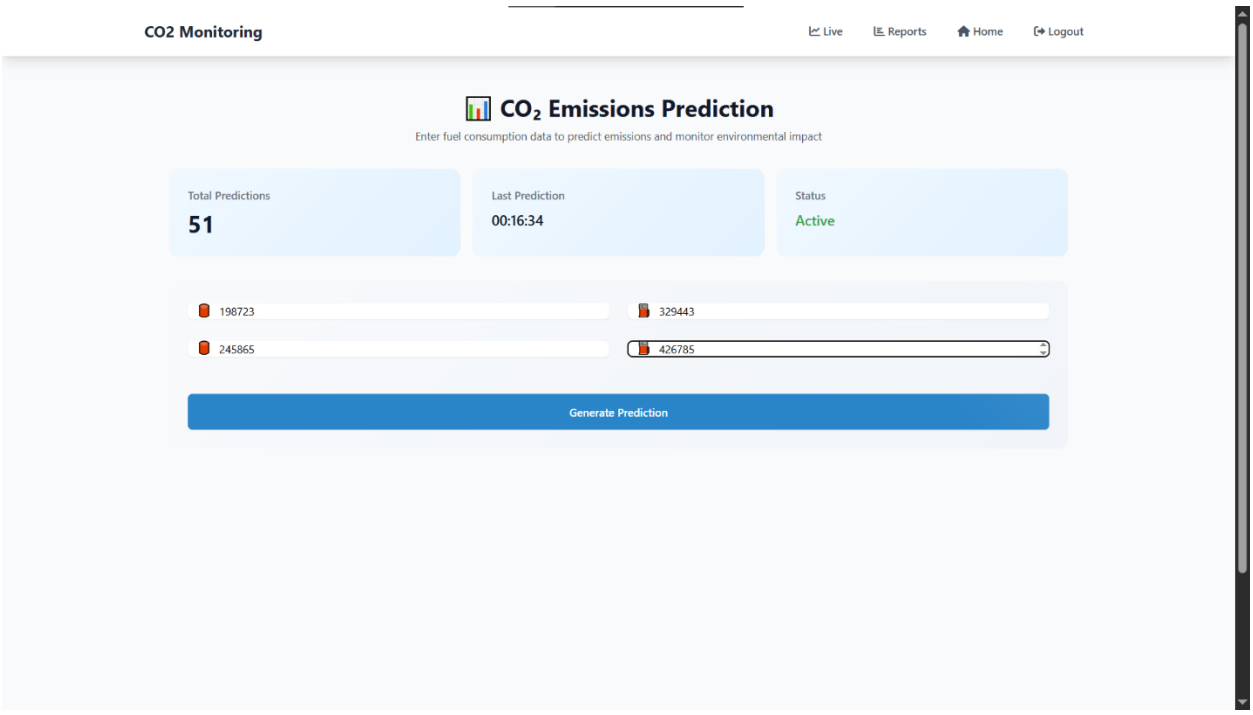


Figure 23 Implemented Emission Prediction Page

This version of the prediction interface is tailored for regulators. It allows them to manually input data for fuel consumption and instantly receive GHG predictions. After entering values for Diesel, Kerosene, Crude Oil, and Gasoline, the page uses our machine learning model to return predicted emissions for CO_2, Methane, Nitrous Oxide, Oil Co2,Gas Co2, and Coal Co2.

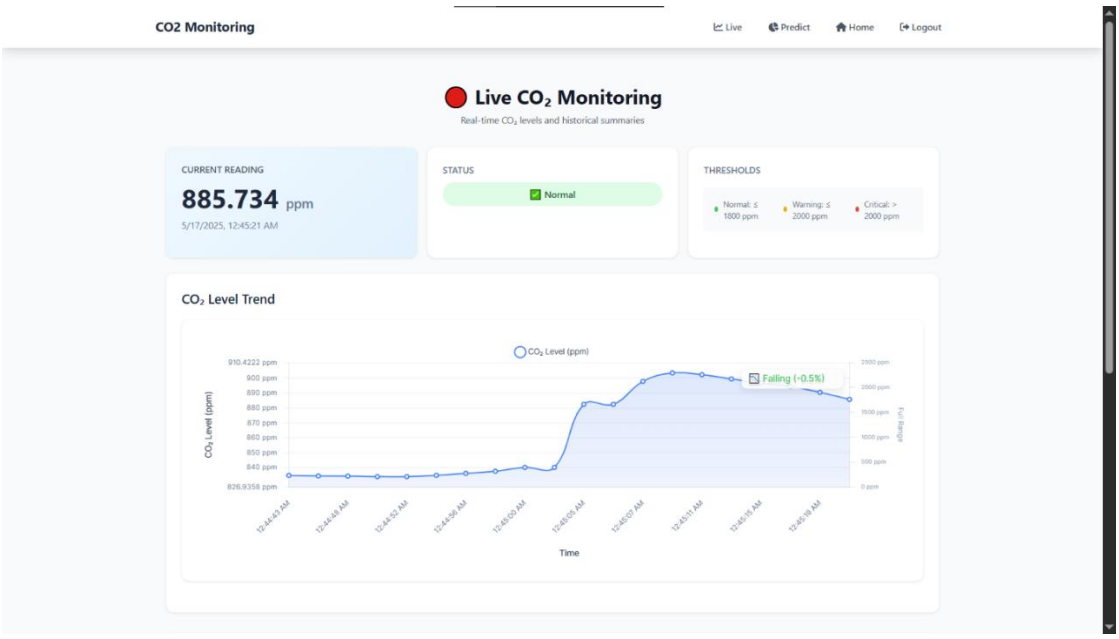


Figure 24 Implemented Live Monitoring page - normal status

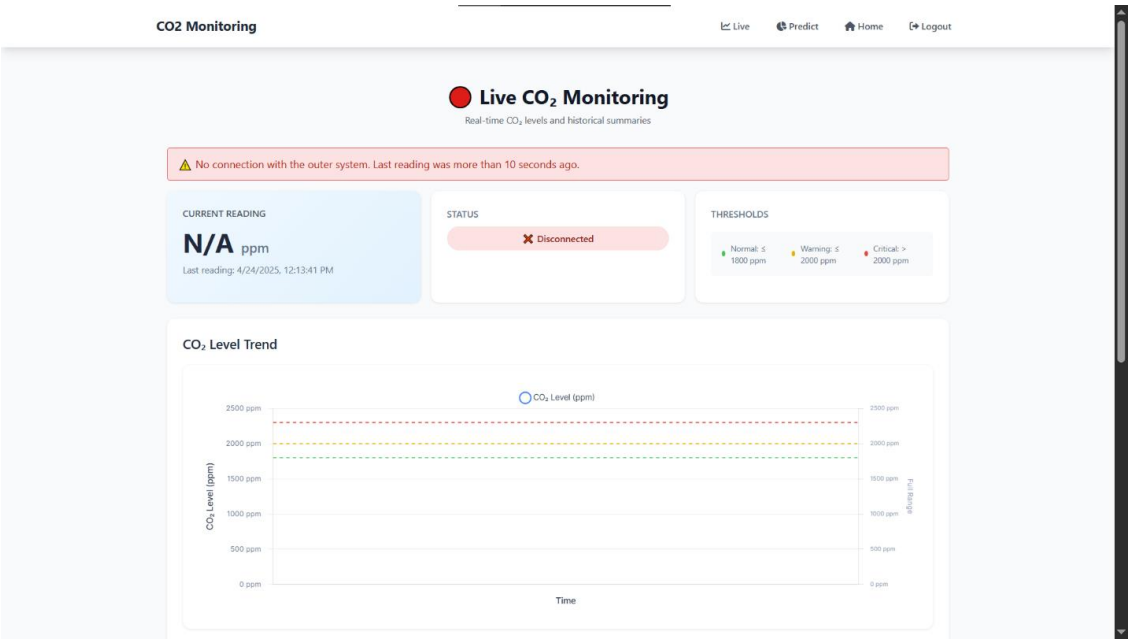
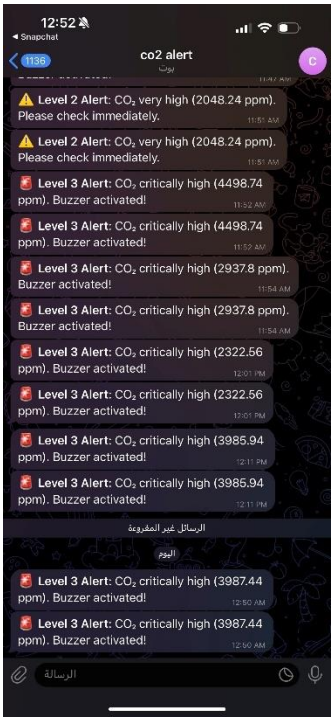


Figure 25 Implemented Live Monitoring Page - Sensors disconnected

Live CO_2 Monitoring Page

Accessible to operators, this page displays the latest CO_2 readings retrieved from the Raspberry Pi and SCD-30 sensor every few seconds. If the sensor connection is lost or delayed, a warning is shown to notify the user. Below the live data, there's a filter section where users can view CO_2 summary values (min, max, average) for any custom time interval.



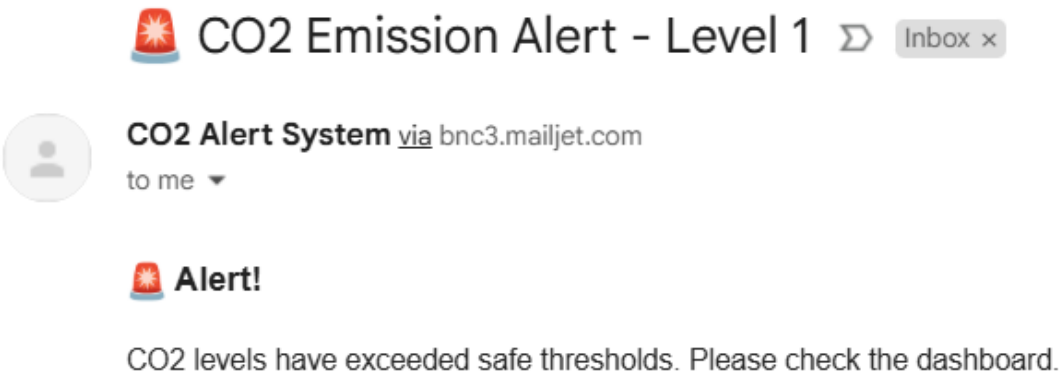


Figure 26 Implemented Live Email Alerts

Figure 27 Implemented Live SMS Alerts

The system implements a multi-channel alert mechanism to ensure timely notification of critical CO_2 levels. When a threshold is exceeded, the system first sends an automated email alert to designated operators, providing an early warning. If CO_2 levels continue to rise, a Telegram message is dispatched for immediate attention. In the event of a severe breach, an active buzzer is triggered on-site to provide an audible alarm, ensuring rapid response even if digital notifications are missed. This layered approach maximizes the chances of prompt intervention and enhances overall safety.

Discussion on the Preliminary Results

The system implementation has demonstrated promising results across all major components, including hardware integration, machine learning models, database infrastructure, and the front-end interface.

The IoT hardware setup, consisting of a Raspberry Pi 5 connected to an SCD-30 sensor, successfully captures real-time CO_2 levels and transmits data to a Docker-hosted MySQL database. This live data acquisition process operates with stable connectivity and minimal latency, forming a reliable input stream for both monitoring and prediction tasks.

On the software side, the role-based web application is fully functional for both Oil Operators and Environmental Regulators. Each user is provided with a customized dashboard. Operators can input operational fuel data, view predictions, and monitor live CO_2 readings, while regulators can upload test metrics, review reports, and trigger alert mechanisms. All system functions, including login, role authentication, data storage, report generation, and alert notifications (via email and SMS), are working as intended.

Regarding the machine learning models, both the **LSTM** and **XGBoost** approaches have been trained and evaluated. The **XGBoost model** achieved outstanding performance, with an R^2 of approximately 0.98 and MAPE below 1%, indicating high prediction accuracy and robustness. It currently serves as the primary model for emission forecasting. The **LSTM model**, while effective in identifying trends and handling temporal dependencies, displayed slightly wider confidence intervals and higher variance in residuals. Nevertheless, it provided valuable insights and remains a strong alternative.

The attempt to ensemble LSTM and XGBoost predictions was postponed due to fundamental architectural and preprocessing differences. As a result, each model is currently maintained independently, with XGBoost integrated into the operational system.

Overall, the preliminary results confirm the feasibility and effectiveness of the proposed solution. The system provides accurate CO_2 forecasts, real-time environmental monitoring, and a responsive user interface—demonstrating strong potential to enhance sustainability practices in petroleum operations.

7 Work Plan

7.1 Project Proposal Work plan

Table 15 Project Proposal Work Plan

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15	W16
Understanding background																
literature review																
Looking for a training dataset																
Defining project scope																
Analyzing project requirements																
Milestone 1 report finalization																
Presentation preparation																
Project analysis																
Finalizing proposed design and project requirement																
Milestone 2 report finalization																
Presentation preparation																

7.2 Project Implementation Work Plan

Table 16 Project Implementation Work Plan

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15	W16
Dataset preprocessing and cleaning																
Initial CNN training																
Building IOT device																
Website development (Framework)																
Testing the initial design and report the initial results																
Analysis of primarily results																
MS3 Report																
MS3 presentation																
Website development (Framework with Database)																
Integrating IOT Device with Web Framework																
Improving/Fine tunning the Model																
System Testing																
MS4 report																
MS4 presentation																

Conclusion

CO_2 emissions represent a significant environmental challenge within the petroleum industry, contributing to climate change and ecological degradation. The proposed system leverages IoT technology and machine learning. Specifically, the XGBoost model. To monitor and accurately predict emissions based on real-time and historical operational data. With a live sensor network and a role-based web platform, the system offers actionable insights for both oil operators and environmental regulators. The result is a user-friendly tool that enhances emissions forecasting and supports informed decision-making. By integrating this solution into petroleum operations, Saudi Arabia can strengthen its sustainability efforts, optimize resource usage, and improve compliance with environmental standards.

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9 Appendix

Table 17 Table of Abbreviations

Abbreviation	full form
XGBoost	Extreme Gradient Boosting
UN	United Nations
SVR	Support Vector Regression
sMAPE	Symmetric Mean Absolute Percentage Error
SDGs	Sustainable Development Goals
SCD-30	Sensor model (Adafruit SCD-30)
SCADA	Supervisory Control and Data Acquisition
RMSE	Root Mean Squared Error
RBAC	Role-Based Access Control
PoE	Power over Ethernet
NDIR	Non-Dispersive Infrared
ML	Machine Learning
MAPE	Mean Absolute Percentage Error
MAE	Mean Absolute Error
LSTM	Long Short-Term Memory
IoT	Internet of Things
HVAC	Heating, Ventilation, and Air Conditioning
GBR	Gradient Boosting Regressor
FFNN	Feedforward Neural Network
ERP	Enterprise Resource Planning
DL	Deep Learning
CO_2	Carbon Dioxide
CNN	Convolutional Neural Network
API	Application Programming Interface

ANN	Artificial Neural Network
MSE	Mean Squared Error
LOOCV	Leave One Out Cross-validation
GHG	GreenHouse Gasses